

**INSTRUMENTAL AND AFFECTIVE MASS MURDER: ESTABLISHING A
PREDICTIVE TYPOLOGY WITH COMPUTER-MEDIATED LINGUISTIC
ANALYSIS**

A dissertation submitted by

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to

FIELDING GRADUATE UNIVERSITY

in partial fulfillment of the
requirements for the degree of

DOCTOR OF PHILOSOPHY IN PSYCHOLOGY

with an emphasis in
Clinical Psychology

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ABSTRACT

This project explores the potential existence of an underlying predictive psychological typology of individuals who commit mass murder. The foundational hypothesis of this paper explores how the linguistic characteristics of pre-attack communications can categorize perpetrators of mass murder as either affective or instrumental and how this categorization may influence attack behaviors. The software program Linguistic Inquiry and Word Count was used to generate linguistic scores for 100 pieces of pre-attack communications. *K*-means cluster analysis of the linguistic scores generated by this program revealed two distinct cluster types: Cluster 1, the seemingly more affectively motivated perpetrator, and Cluster 2, the seemingly more instrumentally motivated perpetrator. Further, binary logistic regression demonstrated that specific attack behaviors predict cluster membership effectively; specifically, the number of victims killed, the perpetrator's use of demographic-based targeting, the presence of the perpetrator's home among the attack locations, and whether an attack ends due to the non-self-inflicted death of the perpetrator.

Keywords: LIWC, mass murder, instrumental violence, offender typology, linguistic analysis

DEDICATION

For my loving family in recognition of their countless sacrifices, unending support, and enduring belief in my ability to reach this important milestone.

ACKNOWLEDGMENTS

Thank you, Dr. Hawkins, for your unparalleled speed, wisdom, and generosity of spirit that has aided me in countless ways. Thank you, Dr. Hanson, for consistently providing me with the grounding and support I sorely needed across these years. Thank you, Dr. Cornelius, for your expertise and willingness to take on this project, despite how incredibly busy and in-demand you are. Double thanks to you, Dr. Boyd, for your contributions to the development of LIWC2015 and your friendliness, good-humored responsiveness, and consistent guidance as one of my committee members.

I must also recognize the work of others that allowed me to complete this research, including the creators and curators of the Stanford Mass Shootings of America data project, the Mother Jones dataset of US mass shootings, and the Gun Violence Archive's dataset of mass shootings. Particular thanks to Dr. Peter Langman and the information he makes available through his website, SchoolShooters.info. I am also deeply appreciative of Dr.s James Pennebaker, Roger Booth, and Martha Francis for their work in the field of linguistic analysis and the development of LIWC2015.

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CHAPTER 1: INTRODUCTION TO THE STUDY

The project described in this paper explores the potential existence of an underlying predictive psychological typology of individuals who commit mass murder and how different offender types may impact attack behaviors. Further, this project intends to show the utility of computer-mediated linguistic analysis to establish this typology and contribute to understanding, preventing, and mitigating this kind of violence. Past investigations of mass murder by social scientists have been limited by the frequent deaths of the perpetrators immediately after the completion of the crimes, prohibiting formal assessment of the offender's underlying motivations and psychological characteristics.

Perpetrators of mass murder frequently create pre-attack communications, and though perpetrators often do not survive their act of violence, their communications do. Computer-mediated linguistic analysis allows for post-offense investigation of the offender's pre-attack motivation and psychological characteristics, regardless of the offender's status. Further, there are strong indications in the literature that linguistic analysis can reveal psychological characteristics of an offender that point to instrumental versus affective motivation. Distinguishing instrumental offenders from affective offenders may support the creation of a predictive, dichotomous typology of mass murderers, meaningfully linking offender and offense characteristics.

The literature suggests that whether an offender is instrumentally or affectively driven may influence how lethal the attack will be, whether the attacker is personally tied to an attack location, and if specific individuals or groups are targeted for violence. Affective and instrumental drivers are potentially measurable by systematically assessing language use by perpetrators before their attack, with factors established in the literature to be psychologically

meaningful; specifically, measures of cognition, self-perception, and emotionality, in addition to pronoun use as a psychometrically robust and empirically validated measure of self-esteem and emotional state. Since instrumental offending is frequently associated with psychopathy in the literature, this study seeks to examine if instrumental offenders can be differentiated from affective offenders by the former's comparatively higher levels of analytical thinking, elevated self-perception, and lower levels of emotionality. This study further investigates if traditionally psychopathic traits, such as resistance to fear, may enhance an instrumental offender's confident, analytical approach, thereby changing how this type of offender completes an attack. Because violence is not an end in and of itself for instrumental offenders, but instead is only a means to another goal, it is posited that these offenders may be less likely to commit suicide upon completion of their crimes. Further, this view of violence as a means may indicate that instrumental offenders are more thoughtful and planful, leading to greater effectiveness (i.e., lethality). Lastly, because instrumental killers are not emotionally driven, they may be less likely to target victims with whom they have a personal relationship and be more likely to kill randomly at a location to which they are not personally connected.

To explore the background of this research area, this paper presents the reader with a literature review on the studied topic, including a discussion of previously proposed typologies for perpetrators of mass violence, existing writings on the differences between instrumental and affective violence, as well as a presentation on the use of computer-mediated linguistic analysis in the field of psychology, specifically the software program Linguistic Inquiry and Word Count (LIWC). Following the literature review, this paper explores previous challenges to the empirical investigation of an underlying predictive typology for perpetrators of mass violence. This exploration focuses on the problems of defining mass murder and advocates for identifying and

analyzing the most impactful discretizing variables related to offender and offense characteristics.

To accomplish the study of the proposed study questions, this project uses publicly available materials for linguistic analysis, including 100 observations of pre-attack communications from perpetrators of mass violence. Following linguistic analysis by the software program LIWC, *k*-means cluster analysis and binary logistic regression tested the proposed hypotheses. The results of these analyses are followed by recommendations focused on potential practical applications. Lastly, the paper presents a discussion of this study's limitations and possible next steps for future research.

CHAPTER 2: LITERATURE REVIEW

Despite mass murder being a relatively rare occurrence worldwide, in the United States (US), the number of mass shootings in the past decade is almost 250% higher than the prior decade, with 57% of all recorded mass shootings occurring within the past 10 years (The Office for Victims of Crime, 2018). The Federal Bureau of Investigations (FBI) reports 250 active shooter incidents between 2000 and 2017 in the US, creating 2,217 casualties, not including the perpetrator (Silver, Simons, & Craun, 2018). When the phenomenon of mass murder is examined internationally and includes instances of mass violence not involving firearms, these figures increase substantially. Even a single life lost to homicide is too many, and the massive and long-term negative psychological impact of mass violence underscores the need for social scientists to leverage criminology, linguistics, and psychometrics to combat this kind of event. The following literature review presents the current body of knowledge for existing typologies for general violent offenders and mass murderers, the psychometric value of linguistic analysis, and methods of computer-mediated linguistic analysis.

Mass Murder Typology

Instances of mass murder, attempted mass murder, and mass murder/suicide are extraordinary forms of violence that differ essentially from other forms of violence. Despite the sensational nature of these crimes, these incidents are not well-understood by social scientists. The formation of typologies is critical for broadening the knowledge base of various behaviors (Collier, LaPorte, & Seawright, 2012), and there have been many historical attempts to establish mass murderer typologies. The vast majority of existing typologies are contingent on one or more of five basic characteristics: location, victim type, motive, weapon, and end result (Bowers, Holmes, & Rhom, 2010). Further, these existing typologies typically exist in one of three

categories: typologies that differentiate mass murderers from the general, non-offending population; typologies that distinguish mass murderers from other types of violent offenders; and typologies that seek to define different kinds of mass murderers.

One of the earliest attempts at mass murderer classification was Dietz's (1986) three-part model, which included *family annihilators*, *pseudo commandos*, and *set-and-run killers*. Holmes and Holmes (1994) added two additional categories: *disciples* and *disgruntled employees*. Kelleher (1997) departed from the use of victim-type and methodology seen in these prior models, focusing instead on offender-motivation as a grouping factor. Kelleher's model consists of seven categories: revenge, perverted love, politics and hate, sexual homicide, mass execution, insanity, and the seemingly unmotivated. More recently, researchers have attempted to generate an empirically-based typing system, but the results of many of these are often conflicting and at odds with what the researchers anticipated and the findings of other similar studies.

The findings of prior studies aiming to create empirically-based groupings for perpetrators of mass violence have resulted in little clarity or consensus. The primary stumbling block for these empirical studies seems to be the identification of meaningful grouping variables. For instance, though age and attack location are reasonable candidates as meaningful grouping variables, Lankford's (2013) comparative, quantitative assessment of suicide terrorists and rampage, workplace, and school shooters found that the differences in age and attack location between these offenders were largely superficial. Additionally, Ferguson, Coulson, and Barnett (2011) found that youthful mass murderers appear to be more similar to adult perpetrators of mass murder than different. Other within-groups categorization approaches, such as determining if specific, psychologically-based identifiers meaningfully divide school shooters, can be seen in Langman's (2009) article describing school shooters as either traumatized, psychotic, or

psychopathic; and Ioannou, Hammon, and Simpson's (2015) work dividing school shooters into the disturbed, the rejected, and the criminal. While these approaches echo some of the more theoretical models first introduced to categorize these offenders and remain interesting conceptually from an *ex post facto* perspective, these types of models add little to the prediction of violent behavior.

Generally speaking, the only typology format that presently holds any predictive potential simply distinguishes mass murderers from the general, non-offending population. Several distinct typologies of this format have been proposed based on diverse environmental, psychological, and social factors (Knoll & Meloy, 2014; Langman, 2009; Newman, Fox, Harding, Mehta, & Roth, 2004; Steffgen, Recchia, & Viechtbauer, 2013; Verlinden, Hersen, & Thomas, 2000). Unfortunately, these typologies only generically address attack likelihood and do not offer more specific information about the kind of attack that is more likely to occur. Discovering predictive factors beyond simply whether a person is at-risk for perpetrating or attempting to perpetrate mass violence can help reveal critical threat assessment and risk management information, such as possible location/victim types, potential lethality, and how an attack is more likely to end.

Instrumental and Affective Violence

Violent behavior has long been described as being either instrumental or not, with the *or not* being termed variously as impulsive, expressive, reactive, and, as used in the current project, affective (Dodge, 1991; Fontaine, 2007; Hart & Dempster, 1997; Kingsbury, Lambert, & Hendrickse, 1997; McEllistrem, 2004; Meloy, 1988). It appears reasonable that mass murder, being one form of violent behavior, could also be distinguished as instrumental or affective. Despite variation in the term used, the distinction between instrumental and affective offending

is two-fold: the level of emotionality associated with the offense and whether violence is the means or the end. With affective offending, the violence is the end, as this type of crime is primarily motivated by the desire to hurt another person; conversely, in cases of instrumental violence, the violence is simply a means of accomplishing another goal. Further, affective violence is a highly emotional act, whereas a hallmark of instrumental violence is the absence of emotionality and the presence of detached calculation.

In the literature, authors frequently associate instrumental offending with psychopathy and assert that the non-emotional, psychopathic offender is more likely to engage in instrumental violence, whereas the emotional, non-psychopathic offender is more likely to engage in affective violence (Glenn & Raine, 2009). These assertions are not unfounded, as one study suggests that 93.3% of the homicides committed by psychopathic offenders are dispassionate, instrumental crimes (Woodworth & Porter, 2002). Research further suggests that when a psychopathic offender commits an instrumental crime, there is typically an absence of emotional arousal, fear, and negative anticipation of punishment (Lykken, 1995; Porter & Woodworth, 2006; Williamson, Hare, & Wong, 1987).

Due to the level of planning and preparation involved in acts of mass murder, there has been a historical assertion that all acts of mass murder are instrumental in nature (National Center for the Analysis of Violent Crime – Behavioral Analysis Unit, 2015); however, closer inspection of offender behavior during an attack suggests that this may not be true. For instance, Declercq, Frédéric, and Audenaert (2011) find relatively wide variation in levels of emotional arousal experienced by perpetrators of mass violence, making it reasonable to suspect that affectively-motivated perpetrators may exist. Further, there are highly conflicting findings in the research with little consensus regarding certain characterological features of mass murderers. For

instance, Thompson and Kyle (2005) suggest that these perpetrators are likely to be suffering from low self-esteem, which would be more typical of an affective offender, while other authors suggest that these perpetrators are more likely to be psychopathic (Palermo, 1997) and/or narcissistic (Hempel & Richards, 1999; Virtanen, 2013), which is more aligned with an instrumental offender-type. Still, other researchers offer evidence of appreciable characterological heterogeneity in this population (Bondü & Scheithauer, 2015; Wike & Fraser, 2009).

Qualitative evaluation of individual cases supports the presence of heterogeneity in this group and effectively challenges the assertion of a universal type of mass murderer. The heterogeneity of this group is particularly observable when considering the crimes of Seung-Hui Cho, who killed 32 people and wounded 17 others at Virginia Polytechnic Institute and State University on April 16th, 2007, and Anders Breivik of Norway, who killed 77 people across two different locations on July 22nd, 2011. The multimedia manifesto created by Cho is far more emotionally charged than the cool detachment present in the 1,500(+)-page manifesto published by Breivik before his attack. Consider the following excerpt from Cho's manifesto seeming to communicate an emotional drive consistent with an affective motivation:

Congratulations. You have succeeded in extinguishing my life. Vandalizing my heart wasn't enough for you. Raping my soul wasn't enough for you. Committing emotional sodomy on me wasn't enough for you.

Now consider some of the titles of the chapters included in Breivik's manifesto:

- *The Movement for Academic Reform*
- *Further Readings on the Frankfurt School*
- *Modern Aftermath of the Crusades*
- *History of the Islamic Ottoman Turkish Empire I (1299-1876)*
- *The EUSSR/USASSR Media Hegemony*
- *US and European Nationalist Rhetorical Differences*

The images each included in their manifestos, as displayed in Figure 1 and Figure 2, helps to illustrate further some meaningful differences between these attackers, their mindset, and the emotionality of their pre-attack communications. In the first picture, Cho is holding a knife to his own throat with his eyes closed, while in the second picture, Breivik smiles toward the camera, wearing a self-made uniform.

This pair of cases is only one example of the repetitious illuminating dichotomy that appears when comparing mass murderers and supports the possibility that these kinds of crimes may be defined meaningfully by motivation type. Additionally, since motivation type may be expected to be accompanied by naturally opposing levels of emotional arousal, fear, and negative anticipation of punishment, it may follow that these level variations may change how lethal an attack is, whether the attacker has a personal connection to the attack location, whether and how victims are targeted, and whether the perpetrator is likely to commit suicide at the conclusion of the event or if the event only concludes when the attacker is killed or apprehended. If this distinction exists among mass murderers, the proposed method of computer-mediated linguistic analysis is an excellent choice for uniformly uncovering and measuring psychometric variables associated with these different psychological states.

Figure 1

Seung-Hui Cho Self-Portrait



Note. Included in the materials distributed by Cho to NBC News prior to his attack.

Figure 2

Anders Breivik Self-Portrait



Note. Included in the web-published manifesto Breivik distributed prior to his attack.

Computer-mediated Linguistic Analysis

When language is acknowledged as a medium for expressing internal thoughts and emotions, detectably influenced by personality and mental health (Tausczik & Pennebaker, 2010), its value as an assessment target is clear. Researchers have persuasively demonstrated that natural language use is integral to the study of thing or person orientation (McIntyre & Graziano, 2016), trait personality (Ireland & Mehl, 2014; Kwantes, Derbentseva, Lam, Vartanian, & Marmurek, 2016), and state inhibition, activation, and motivation (Cohen, Minor, Baillie, & Dahir, 2007). Given how few perpetrators of violence survive beyond their offense, language assessment is even more valuable for the current project, as it allows for posthumous assessment. Though deceased offenders cannot be assessed directly, as many as 67% of perpetrators of mass violence create meaningful pre-attack communications (Hempel, Meloy, & Richards, 1999; Langman, 2013; Meloy, Hempel, Mohandie, Shiva, & Gray, 2001; Silver, Horgan, & Gill, 2018). Subjecting these pre-attack communications to computer-mediated linguistic analysis offers an opportunity for objective, standardized, and quantitative exploration of language use in this population regardless of the perpetrator's status at the time of analysis.

Tausczik and Pennebaker (2010) described the general psychometric value of computer-mediated linguistic analysis in great detail. One highly relevant example of the usefulness of computer-mediated linguistic analysis for the present study is the work completed by Neuman, Assaf, Cohen, and Knoll (2015). These researchers used vector space models of semantics to compare the texts written by six school shooters to a control corpus. These models differentiated offender texts from the control corpus, with offender texts showing substantially higher scores for indications of narcissism, humiliation, and vengeance. This kind of statistical differentiation helps to support the notion that computer-mediated linguistic analysis can differentiate attackers

from the general population. This work also supports the idea that if sufficient variation exists among pre-attack communications, it may also be possible to accomplish a within-groups differentiation of attackers.

Linguistic Inquiry and Word Count.

A literature review reveals that Linguistic Inquiry and Word Count (LIWC; Pennebaker, Booth, Boyd, & Francis, 2015) is one of the more popular methods among researchers for conducting computer-mediated linguistic analysis. LIWC is an empirically validated tool for measuring linguistic expressions of emotion. This program quickly and systematically transforms qualitative language data into quantifiably analyzable data (Kahn, Tobin, Massey, & Anderson, 2007). Examination of the multiple studies using LIWC suggests that its popularity is due to the diverse and powerful applicability of this tool in psychological research focused on the psychometric value of language.

Some examples of studies using LIWC include its use as a predictive measure to determine future college student performance (Robinson, Navea, & Ickes, 2013) and recidivism among sexual offenders (Drouin, Boyd, & Greidanus-Romaneli, 2018). It has also been used discriminatively to identify psychological markers of political ideology (Robinson, Boyd, & Fetterman, 2014), as well as to distinguish lone actor terrorists (Baele, 2017) and psychiatric patients (Junghaenel, Smyth, & Santner, 2008) from the general population. It has additionally been used to measure well-established psychological constructs such as depression (Newell, McCoy, Newman, Wellman, & Gardner, 2018), psychopathy (Hancock, Woodworth, & Porter, 2013), and suicidality (Handelman & Lester, 2007; Lester, 2009; Lester 2014). It has also been used to measure Big Five personality traits (Hirsh & Peterson, 2009), such as extraversion and introversion (Beukeboom, Tanis, & Vermeulen, 2013). It has even been used to discover how an

individual's use of language changes when describing dreams (Bulkeley & Graves, 2018; Hawkins & Boyd, 2017).

More specific to the current project's focus, LIWC has been used to discover the increased presence of communicated drives for power in the manifestos of perpetrators of mass murder when compared to control corpora (Leffew, 2017). Additionally, researchers have used LIWC to analyze general student writings, suicide notes, and the pre-attack communications of mass murderers (Egnoto & Griffin, 2016). These researchers found LIWC to be an effective tool in identifying suicidal and homicidal writings as independently distinct. These findings prompted the authors to advocate for the investigation of using computer-mediated linguistic analysis to detect harmful ideation.

In summation, the existing proposed within-groups typologies of mass murderers offer little to no predictive value regarding specific attack behaviors. Creating a typology that offers predictive value requires the linkage of a typology to a set of pre-attack behaviors. The lack of contemporary progress in establishing a predictive within-groups typology may indicate an erroneous presumption that all mass murderers are instrumental offenders. The literature suggests that there may be a missed within-groups typology distinguishing affective perpetrators from instrumental perpetrators. This missed typology could permit the predictive linking of offender type with specific attack behaviors that has so far been missing. Additionally, computer-mediated linguistic analysis, specifically LIWC, seems capable of uncovering such a typology, with a substantial benefit of this approach being the ability to complete post-event psychological evaluation of attackers, regardless of the perpetrator's status at the time of analysis.

CHAPTER 3: STATEMENT OF THE PROBLEM

The development of practical prevention efforts against future instances of mass violence seems to be presently impeded by an incomplete understanding of this complex phenomenon. When considering the professional and scientific responsibilities of psychologists to society, there is a demand for social scientists to promote public safety through the development of evidence-based prevention efforts. Gaining a deeper and more accurate understanding of these horrific events is the only reasonable path to accomplishing this goal successfully. Unfortunately, a high level of empirical and theoretical conflict in the literature challenges gaining this kind of understanding. These conflicts could be the result of a missing within-group typology of offenders that considers how affective and instrumental motivations may meaningfully impact attack behaviors. The present project seeks to investigate the potential presence of such an underlying typology.

Past Challenges to Empirical Investigation

The literature review that was completed in preparation for the current study revealed that mass murder has received substantially less attention from researchers and law enforcement agencies than other forms of multicide, such as serial murder. Curiosity is the first necessary component to generating the interest that leads to funding and high-quality investigation. Compared to other types of violent offending, there seems to be a historical lack of equitable interest in mass violence. This neglect may be due to multiple factors.

Fox and Levin (2003) outlined that mass murder, unlike serial murder, does not require prolonged attention from law enforcement because the perpetrator is typically neutralized (i.e., killed, found dead, or taken into custody) shortly after coming to the attention of law enforcement and therefore does not present an ongoing threat. Further, since offenders often do

not survive the event, they cannot be subjected to after-event assessment for motivation and defining psychological characteristics, which may partially explain the lack of interest from behavioral scientists. Compounding this seeming lack of interest is that the little research that has been done seems to suffer from phenomenological definitions that are conflicting, inaccurate, or problematically exclusionary or inclusionary. There further seems to have been a historical focus on non-discretizing variables without robust predictive function. This project aims to remedy all these obstacles to empirical study.

Defining mass murder. Despite the notoriety of mass murder, this type of violent behavior is not well understood in the behavioral sciences, and empirical investigation suffers from weak definitions, limiting the development of effective, empirically-based prevention efforts (Bowers et al., 2010). Many researchers use the FBI's definition of mass murder, which states that mass murder is the killing of four or more individuals without a significant cooling-off period (2005), with no specificity as to what makes a cooling-off period significant. The vagueness of this definition and the inclusion of a possibly externally sourced success component that is not necessarily psychologically rooted may be causing researchers to be overly exclusionary in case selection.

The actions of Geddy Kramer effectively challenge the use of a success component as a defining characteristic. The FBI would not define Kramer as a mass murderer, despite his possession of the typical features of a mass murderer and his apparent preparation and commitment to complete a highly successful attack. Shortly before he shot six of his coworkers at the FedEx facility where he worked, Kramer gathered his equipment, which included a Mossberg Model 930 semi-automatic shotgun, four Molotov cocktails, a knife, at least 50 shotgun shells, a bandolier to carry the shells, and a pouch for the Molotov cocktails (Judd,

2014). Of all those shot by Kramer, even those at close range, the only person he successfully killed was himself. Multiple factors seemed to have influenced the survival of all of Kramer's victims, including efficient emergency response, rapid access to medical care, and Kramer's naïve choice of a less-lethal type of ammunition. The survival of his victims does not seem to be due to him being remarkably psychologically different from those who do fulfill the FBI's limited definition of mass murder. Cases such as these prompted the development of a new definition for use in this study that seems less susceptible to the influence of external factors than the FBI's. Though a researcher using the FBI's definition would need to exclude Kramer, this study's definition allows Kramer to be included and treats his lack of success as a typology factor.

Additionally, the historical tendency to differentiate lone actor terrorists from other mass murders may be misguided. Wellington Menezes de Oliveira, who killed 12 elementary school children and wounded 12 others in 2011, is an excellent example of an individual often viewed as a lone-actor terrorist and not a mass murderer. When his pre-attack communications are examined closely alongside the other characteristics of his offense, there is little substantive difference between him, his circumstances, and mass murderers who are not considered lone-actor terrorists. Though he appears to have a psychological profile more traditionally associated with mass murder than political terrorism, his multiple references to Islam in his pre-attack communication may be why he is more often considered a terrorist. This study only excludes individuals as terrorists if there is a known affiliation with a recognized terroristic organization and the organization provided aid and support in preparation for and perpetration of, the attack.

The definition used by this study defines a mass murderer as any perpetrator who presents as prepared and motivated to murder four or more individuals, with *insubstantial time*

passing between violent acts. Insubstantial time is considered any period less than that which would be required to plainly identify subsequent acts of violence as separate, distinct, and unrelated to the preceding act of violence. The number and time frame referenced in the definition is required to distinguish mass murder from single victim events, spree murder, and serial murder. Further, to avoid confounding mass murderers with the distinctly separate category of perpetrators known as *family annihilators*, this study excludes those whose only victims were their family members.

Variable selection. The present study's focus is identifying potentially predictive variables, specifically how offender characteristics may influence offense characteristics. Meta-analysis on the profiles of homicidal offenders supports a specificity hypothesis of stable relationships between offense behaviors and offender characteristics (Crabbé, Decoene, & Vertommen, 2008), and it is reasonable to anticipate that similar stability would be observable among mass murderers. As posited in the introduction to the current study, there is reason to suspect that mass murder may be instrumentally or affectively motivated. If true, instrumentality and affectivity should manifest in the psychological characteristics of the offender and the specifics of the attack. As noted previously, many perpetrators do not survive their attack, limiting the ability of researchers to subject perpetrators to formal psychological examination. Further, any *post hoc* assessment of an attacker would likely be delayed, and as a result, its ability to accurately capture an attacker's emotional state immediately preceding the perpetration of violence would be considerably limited. Linguistic analysis of pre-attack communications side-steps these barriers and permits pre-attack psychological characteristics to be measured and analyzed as potentially meaningful predictors of attack behaviors, specifically: the lethality of

the attack, the offender's connection to the attack location, whether and how the offender dies at the conclusion of the crime, in addition to if and how the perpetrator engages in victim selection.

Offense characteristics. As this study is intended to provide actionable information to promote the practice of accurate and effective threat assessment and risk management in the domain of public safety, it is important to consider how the results of the current study support this goal. Firstly, it must be understood that the proposed use of linguistic analysis to determine offender types is not intended to provide a between-group comparison but rather only a within-group comparison. In other terms, this study is not intended to provide predictive information about whether an attack will occur, but rather, if an attack were to occur, what type of attack is more likely based on measurable psychological variables within a suspected pre-attack communication.

Because the process of threat assessment risk management is centered on an equation evaluating the probability an event will occur (i.e., the threat) and the potential impacts of the event (i.e., the risk), this process is heavily impacted by the characteristics of a potential offense. Both sides of this equation are used to determine a level of response, and the demand for intervention for low probability events increases alongside the possible impacts of the event, as is illustrated in Figure 3.

One example of a low-threat/low-risk event is the burglary of an unoccupied home. For most individuals, the likelihood of having their home burglarized is relatively low, as are the potential impacts of the burglary. This assessment causes many to leave their homes unguarded regularly, relying simply on locked doors to protect the assets inside. As we move along the matrix, school shootings are typically considered to be low-threat/high-risk events. While the consequences may be dire, the threat is still at a low enough level that the majority of parents

Figure 3

Risk/Threat Matrix

		Risk →				
Threat ↑		Negligible	Minor	Moderate	Significant	Severe
	Very Likely	Low	Moderate	High	High	High
	Likely	Low	Moderate	Moderate	High	High
	Possible	Low	Low	Moderate	Moderate	High
	Unlikely	Low	Low	Moderate	Moderate	Moderate
	Very Unlikely	Low	Low	Low	Moderate	Moderate

continue to send their children to school every day. At the end of the matrix is a high-threat/high-risk event. An example of such an event is a well-armed individual who has communicated credible intent to commit a mass shooting at an elementary school accessible to the threatener. This scenario is markedly different from the standard low-threat/critical-risk equation of a school shooting, as in this case, there is a known individual with the means and ability to carry out a specific threat. The critical level of both threat and risk presented by such an event would require immediate response beyond the school's standard security protocols.

Since the current study does not address the probability of an event occurring but instead looks at what might happen if the event were to occur, this study's findings are most impactful for the risk management side of the equation. Since research supports the connection between a mass murderer's motivation and the number of people killed and wounded in an attack (Lankford, 2016), one focus of this study is the offense characteristic of lethality. Specifically,

this study seeks to determine whether and how lethality is impacted by affective and instrumental offender motivation. Ideally, the findings of this study would add another level of understanding for the scenario mentioned above of the individual planning to attack an elementary school. Based on the linguistic characteristics of the communicated threat, responding professionals may be armed with knowledge about how deadly the attack is likely to be and can tailor responses accordingly.

However, it is not enough to merely consider how deadly a potential attack may be. Effective prevention also requires anticipating where an attack may occur and who the victims might be. One assessable offense characteristic to aid in this determination is whether the attacker is likely to have a personal connection to the location where the attack occurs. Since instrumental offending is differentiated from affective offending by the relative level of emotionality, it is reasonable to anticipate that affective offenders are more likely to have existing personal connections to their chosen attack locations. Additionally, prior research indicates that offenders who have a personal connection to the attack location generate comparatively lower casualty numbers (Lankford, 2013).

An additional offense characteristic worthy of investigation is whether and how attackers specifically target their victims. Fox, Brook, Stratton, and Hanlon (2016) asserted the importance of victim relationship in the sub-typing of mass murderers, and Petee, Padgett, and York (1997) contend that the target selections made by offenders is highly influenced by an attacker's level of anger and desire for revenge. These authors assert that angry and vengeful killers target specific people and specific places, whereas diffuse targeting occurs in the absence of these states. Further, Mullen (2004) argues that mass murder is most meaningfully differentiated by whether an attacker's victims are targeted based on a personal relationship, group membership, or are

chosen purely at random. He adds that personal agendas, offender-specific social situations, and psychopathologies highly determine patterns of victim selection. The work of these researchers strongly support a line of inquiry into how instrumental and affective offenders differ in their targeting of victims and choice of attack location.

Finally, it is crucial for those responding to an attack in progress to know if the attack will be ended by the offender's suicide or if the attack will continue until the attacker is killed or apprehended. General research on instrumental offenses, particularly those committed by psychopathic individuals, suggests that this type of attacker would be less likely to be suicidal, as suicide is associated with a level of emotionality that would be more expected of an affectively motivated offender. Further, if the violence itself is the end goal, as would be true for an affective offender, there would be no need to survive beyond the attack; however, if violence is simply a means to another end, as would be true for an instrumental offender, then it would seem that these offenders would be more motivated to survive their attack. Mullen's (2004) research with surviving mass murderers found them to have personalities marked by grandiosity. While it is not possible to tell if this is a feature unique only to surviving mass murderers, it at least supports the possibility that the grandiose, psychopathic, instrumental offender is the attacker least likely to commit suicide at the scene of the attack and supports the investigation of this possibility in the current project.

Offender characteristics. Because the hypotheses of this project suggest that offense characteristics have statistically significant ties with offender characteristics, appropriate offender variables must also be identified for analysis. The current project used LIWC to uncover language-based psychometric indicators of affective or instrumental tendencies in those who have attempted or succeeded in committing an act of mass violence. The chosen LIWC

generated variables for this study are named *summary variables*. Summary variables are comprised of the composites of other LIWC scores, such as pronoun use.

Pronouns are parts of speech often ignored in more traditional, content-driven linguistic analyses, despite these smaller function words being important markers of emotion, self-perception, and cognition (Pennebaker, Mehl, & Niederhoffer, 2003). The increased use of first-person personal pronouns has been associated with depression (Bernard, Baddeley, Rodriguez, & Burke, 2016; Himmelstein, Barb, Finlayson, & Young, 2018; Newell et al., 2018), suicidality (O’dea, Larsen, Batterham, Calear, & Christensen, 2017), physical and emotional pain (Tausczik & Pennebaker, 2010), and general negative emotionality (Tackman et al., 2018), in a way that is not moderated by demographic factors (Holtzman, 2017). Increased use of first-person singular pronouns is associated with lower levels of social status and dominance (Arguello et al., 2006; Danescu-Niculescu-Mizil, West, Jurafsky, Leskovec, & Potts, 2013; DeWall, Buffardi, Bonser, & Campbell, 2011; Kacewicz, Pennebaker, Davis, Jeon, & Graesser, 2014; Pennebaker, 2011a), and as a measure narcissism (Carey et al., 2015), with narcissism being counter-intuitively negatively associated with increased first-person singular pronoun use (Carey et al., 2015; Jones, Norville, & Wright, 2017). Pronoun use as measured by LIWC has been used previously to identify suicidal individuals (Egnoto & Griffin, 2016; O’dea et al., 2017) and determine group propensity for violence (Pennebaker, 2011b).

By combining the examination of multiple LIWC variables assessing both function words, such as pronouns, and content words, LIWC’s summary variables effectively capture differences in cognition, self-perception, and emotionality. LIWC’s summary variables are *analytical thinking*, *clout*, *authenticity*, and *emotional tone*. These variables were selected because face validity suggests that they will have opposing values for affective and instrumental

offenders. Presented below are the description of these variables from the operator's manual (Pennebaker et al., 2015):

- Analytical thinking - a high number reflects formal, logical, and hierarchical thinking; lower numbers reflect more informal, personal, here-and-now, and narrative thinking.
- Clout - a high number suggests that the author is speaking from the perspective of high expertise and is confident; low Clout numbers suggest a more tentative, humble, even anxious style.
- Authentic - higher numbers are associated with a more honest, personal, and disclosing text; lower numbers suggest a more guarded, distanced form of discourse.
- Emotional tone - a high number is associated with a more positive, upbeat style; a low number reveals greater anxiety, sadness, or hostility. A number around 50 suggests either a lack of emotionality or ambivalence.

The Need for Predictive Metrics

Mass murder is an extraordinarily complex phenomenon, deserving of empirical study by those qualified to investigate the biopsychosocial factors contributing to these instances. One notable reason supporting the involvement of social scientists in mitigating this phenomenon is that history has shown the criminal justice system to be highly limited in the prevention of violence (Felson & Boba, 2010) and legislative responses to violence to be largely ineffective (Brezina & Wright, 2000). For instance, school killings are increasing globally, even in countries where firearm access is highly regulated (Auxemery, 2015). It may even be reasonable to argue that Nikolas Cruz, the man who shot and killed 17 students at a high school in South Florida in 2018, was paradoxically enabled by policies aimed at responding more appropriately to students exhibiting behavioral problems (Richardson, 2018).

Despite pervasive popular beliefs that mass murderers “snap” and that these acts of violence are senseless and random, even cursory examination of the phenomenon quickly dispels these assertions. Governmental studies on instances of mass violence find that most perpetrators display identifiable behaviors, reflecting potentially violent intent, in advance of committing these crimes (Fein, Vosskuil, & Holden, 1998; U.S. Secret Service & U.S. Department of Education, 2004). The FBI’s study of pre-attack behaviors of the perpetrators of mass shootings (Silver, Simons, & Craun, 2018) adds further evidence to dispute these claims. The researchers found that of the 63 perpetrators examined in the study, the majority of subjects spent a week or longer on both planning and preparing for the attack, creating a critical window for identification and response. Additionally, 88% of active shooters, aged 17 and younger, communicated their intent to commit a violent act, and nearly a third of all perpetrators created communications claiming credit and articulating motives for the attack.

The pathway to violence proposed by Calhoun and Weston (2003) outlines five steps that often occur prior to an act of mass violence: grievance, ideation, research and planning, preparation, and breach. This pathway represents multiple stages where evaluation and intervention could occur. Warning behaviors nearly universally precede acts of mass violence and are crucial in the practice of threat assessment and risk management, with indirectly and directly communicated threats representing sufficient justification for operational response (Meloy, Hoffman, Guldemann, & James, 2012).

Unfortunately, those in a position to identify and intervene with attackers before the event are currently without an empirically based approach to objectively evaluate the risk and threat levels a discovered pre-attack communication poses. Research has found that objective threat assessment measures could effectively prevent instances of mass violence, but correctly

analyzing the contents of threats is highly problematic (Capellan & Lewandowski, 2019). The present study, which seeks to use software designed to examine the psychometric properties of language, is a potential first step in developing an objective and empirically supported screening tool for concerning communications.

Research Question

Is there an underlying typology of perpetrators of mass violence, discoverable through linguistic markers, that are related to the attack behaviors of lethality, personal connection to the attack location, targeting of individuals, and whether and how the perpetrator dies at the conclusion of the attack? The research sought to test the following hypotheses:

Hypotheses

- *H₁*: Cluster analysis of summary language variable scores derived from computer-mediated linguistic analysis will suggest the presence of two distinct types among perpetrators of mass murder.
- *H₂*: Cluster membership based on summary language variable scores will have a statistically significant relationship with number of victims killed.
- *H₃*: Cluster membership based on summary language variable scores will have a statistically significant relationship with whether an offender has a legitimate connection to the attack location.
- *H₄*: Cluster membership based on summary language variable scores will have a statistically significant relationship with whether victims were targeted or killed at random.

- H_5 : Cluster membership based on summary language variable scores will have a statistically significant relationship with if the perpetrator was killed by another person at the conclusion of the attack.
- H_6 : Cluster membership can be modeled using binary logistic regression based on the attack characteristics of the number of victims killed, personal connection to the attack location, whether the perpetrator engaged in victim targeting, and whether the attack ended with the perpetrator being killed.

CHAPTER 4: METHODS

Participants

The analyzed dataset contains 100 observations of language generated by individuals who perpetrated an instance of mass violence or attempted and failed to commit an act of mass violence due to circumstances beyond their control. Eliminating the requirement for specific victim numbers to be met for inclusion in analysis avoids some of the definitional problems that have plagued past research efforts. Since there is typically little phenomenological or psychological distinction between those classically characterized as mass murderers and lone-actor terrorists, there are also individuals included who may be more traditionally classified as lone-actor terrorists.

The observations in the dataset include individuals ranging from 12 years of age to 66 years of age (*Mean* = 22.81, *Md* = 20.00, *Mo* = 16.00, *SD* = 9.25). The oldest instance in this dataset is a case from 1966. In this case, after murdering his mother and wife, Charles Whitman traveled to the University of Texas at Austin, where he fatally shot 15 people and wounded 31 others. The most recent instance in this dataset is a case from 2019. In this case, Brenton Tarrant attacked two mosques in Christchurch, New Zealand, shooting and killing 50 people and wounding 50 others.

Demographically, the authors of the sampled texts are overwhelmingly young, White, male, and residents of the US. There was no purposeful selection or exclusion based upon any of these demographic variables, and the discrepancy between the demographic features of this offender group and the general population is believed to be natural. Inclusion in the sample group relies solely on the commission of mass violence or having a highly obvious intention to commit such an act and the availability of pre-attack communications. The dataset also includes individuals who committed mass violence crimes outside of the US, if they created pre-attack

communications in English, or if a professionally translated version of the communication was available for analysis.

Materials

The analyzed corpora were obtained through publicly available sources. For instance, Dr. Peter Langman, Ph.D., has made an extensive database full of writings of school shooters available through his website, SchoolShooters.info. Other communications were accessed through social media platforms, like Elliot Rodger's video manifesto that he uploaded to the media-sharing site, YouTube. The specifics of each incident were gathered through three primary sources: The Stanford Mass Shootings of America data project (Stanford University, N.D.), Mother Jones' dataset of US mass shootings (Follman, Aronsen, & Pan, 2019), and the Gun Violence Archive's dataset of mass shootings (2019), all of which are available for public use. Since these three databases are dedicated to tracking mass violence committed solely with firearms, additional news media sources were used to gain the particulars of acts of mass violence not involving firearms.

Procedures

Computer-mediated linguistic analysis. This project used the software Linguistic Inquiry and Word Count 2015 (LIWC2015; Pennebaker et al, 2015). LIWC2015 generated scores based on the summary variables of analytical thinking, clout, authenticity, and emotional tone. The summary variables are derived from previously published laboratory findings of LIWC's developers and converted to percentiles based on standardized scores from large comparison samples (Pennebaker et al., 2015). The deriving algorithms for these variables involve normalizing scores based on composites of other LIWC measures. For instance, analytical thinking is derived from the use of articles, prepositions, pronouns, auxiliary verbs,

conjunctions, and negations, while clout is based on certain pronouns, social words, swear words, negations, and differentiation words (Jordan, Sterling, Pennebaker, & Boyd, 2019).

Data preparation. Each language sample needed to be placed into a format analyzable by LIWC2015. Therefore, all materials that were handwritten, graphically captured, or originally in an audio/video format were transcribed into a uniform digital text format. Though LIWC2015 does not require that grammar, capitalization, or sentence structure be corrected, significant misspellings were corrected according to the instructions present in the Operator's Manual (Pennebaker et al., 2015).

Data analysis. The proposed analysis was complicated by a relatively small sample size and the need to account for categorical and continuous variables among the six listed hypotheses. A combination of statistical clustering, non-parametric tests, and logistic modeling was used to overcome these challenges. These tests include *k*-means cluster analysis, the two-tailed Mann-Whitney *U* test, Fisher's exact test, and binary logistic regression.

K-means cluster analysis. The first hypothesis, that cluster analysis of summary language variable scores derived from computer-mediated linguistic analysis will suggest the presence of two distinct types among perpetrators of mass murder, was tested using *k*-means cluster analysis. Cluster membership was determined by the summary variable scores generated by LIWC2015. *K*-means clustering offers several advantages over hierarchical clustering techniques, especially when the number of clusters is known and few (Bayne, Beauchamp, Begovich, & Kane, 1980). First, the method goes through a predetermined amount of iterations of the data to ensure that all cases are assigned to the best cluster (Rencher, 2002). Hierarchical techniques start with one essentially random case as a seed for the first cluster. It is possible that this randomly chosen case may be an outlier, which can then artificially skew the clustering

process (Lange, Iverson, Senior, & Chelune, 2002). The numerous iterations used in *k*-means clustering allow this problem to be corrected, should it occur.

Hierarchical clustering techniques also suffer from a pragmatic disadvantage in that distinct and meaningful clusters may not easily form (Lange et al., 2002). Clusters may be disproportionately large or small, and there is some disagreement in how the actual number of clusters is chosen once the process is complete. In *k*-means cluster analysis, the number of clusters is specified at the outset, which circumvents these difficulties in determining the correct number of clusters (Rencher, 2002); however, this advantage also represents the main criticism of this technique. The number of clusters specified may not be the true number of clusters present in the population. For this research, *a priori* determination of the number of clusters was not seen as a significant disadvantage because the literature supports the presence of two distinct types of violent offenders. The purpose of this study was not to determine the true number of clusters but instead to determine whether two distinct psychological types are detectable and influential on offender behavior.

A final advantage of the use of *k*-means clustering is flexibility in sample size. There appears to be little concern for ensuring adequate sample sizes among the empirical studies in the literature making use of this technique. When constructing clusters, a high number of contributing variables can create a substantial degree of dimensionality, limiting the meaningfulness of any generated cluster. To avoid dimensionality problems, *k*-means clustering should be approached similarly to latent class analysis, where adequate sample sizes are determined by critically evaluating the number of contributing variables. Despite there being no clear “rule of thumb” regarding sample size minimums for *k*-means clustering (Dolnicar, 2002), this study exceeded the standard suggested by Formann (1984) concerning latent class analysis.

For conducting latent class analysis, Formann suggests that the minimum sample size should include no less than 2^k cases (k = number of variables), preferably $5(2^k)$. In this study, there are four summary variables ($k = 4$), meaning that the 100 observations in the data set meet the preferable minimum requirement of 40 to avoid problematic dimensionality and meaningless cluster membership. Lastly, the generated clusters were qualitatively evaluated for theoretical soundness.

Mann-Whitney U. The second hypothesis, that cluster membership based on summary language variable scores will have a statistically significant relationship with the number of victims killed, was tested using the two-tailed Mann-Whitney U two-sample rank-sum test as an alternative to the independent samples t -test. The Mann-Whitney U , also known as the Whitney Wilcoxon test or the Wilcoxon rank-sum test, is non-parametric and has none of the assumptions required for an independent samples t -test, such as normality. Though the distribution of victims killed share a similar shape in the two clusters, the number of victims killed is not normally distributed. Without having observable normal distribution in the data, the Mann-Whitney U was determined to be the most appropriate analysis to compare the differences between the two clusters.

The Mann-Whitney U test compares the number of times a score from one sample is ranked higher than a score from another sample. The scores from both samples are ranked together, with the lowest score being ranked as a 1, and 2 being used for the next lowest score, and so on. When scores have the same value, the scores are given the same ranking. Once all the scores have been ranked, those ranks are added together and then divided by the number of scores. The U statistic is then calculated. The calculated statistic is then compared to the critical U value needed for statistical significance, or if the sample size is large enough to approximate a

normal distribution from the determined ranks, the z-score is used. For this project, an alpha of .05 was used to determine the presence of significant differences in the number of victims killed between the two clusters.

Fisher's exact test. Fisher's exact test were conducted to test H_3 through H_5 . These hypotheses posit that cluster membership based on summary language variable scores will have a statistically significant relationship with whether an offender has a legitimate connection to attack location (H_3), whether specific individuals or groups are targeted (H_4), and if the perpetrator is killed by another person at the conclusion of the attack (H_5). Fisher's exact test is the appropriate statistical test when the purpose of the research is to examine the relationship between two categorical level variables. Contrary to the cell size requirements for the Chi-square test of independence, Fisher's exact test does not have any assumptions; for this reason, Fisher's exact test is a common alternative to the Chi-square test of independence when there are small values in some of the cells of the contingency table (Mehta & Patel, 1983). Fisher's exact test uses a contingency table to calculate the exact p -value, with an alpha of .05, to assess statistical significance while also generating an odds ratio (OR) for cluster membership based on the assessed variable.

Binary logistic regression. The sixth hypothesis was tested using binary logistic regression. This hypothesis posits that cluster membership can be modeled based on the attack characteristics of the number of victims killed, personal connection to the attack location, whether the perpetrator engaged in victim targeting, and whether the attack ended with the perpetrator being killed. Binary logistic regression is the appropriate when the purpose of a study is to determine if a combination of variables can effectively predict a dichotomous dependent variable. In this study, specific attack behaviors (location type, targeting, lethality, and

perpetrator death) are evaluated for their ability to predict the dichotomous variable of cluster membership. Binary logistic regression overcomes many of the restrictive assumptions of linear regression. For instance, normality and homoscedasticity of the residuals are not assumed, and binary logistic regression does not require the complete absence of multicollinearity in the predictor variables.

Though binary logistic regression does not require the complete absence of multicollinearity, this is not to say that there are no limits to acceptable levels of multicollinearity. For this study, variance inflation factors (VIF) were calculated to determine the presence of problematic multicollinearity. Further, the overall significance of the model was examined using the X^2 omnibus test of model coefficients, and McFadden's R^2 was calculated to estimate the percentage of variance accounted for by the predictor variables. The OR was also calculated for each included variable to evaluate how strongly each variable predicts cluster membership.

CHAPTER 5: RESULTS

Hypothesis 1: Summary Variable Clusters

The first hypothesis tested was that cluster analysis of summary language variable scores derived from computer-mediated linguistic analysis would reveal the presence of two distinct types among perpetrators of mass murder. This hypothesis was tested with *k*-means cluster analysis. First, the number of clusters, in this instance two, was determined through the *a priori* process described previously. Next, the *k*-means cluster analysis was conducted and assessed through descriptive statistics and linear discriminant analysis (LDA). For each cluster group, means (*M*), standard deviations (*SD*), and sample size (*n*) are shown in Table 1.

Table 1

Summary Statistics Table for Each Summary Variable Grouped by Cluster

Variable	<i>M</i>	<i>SD</i>	<i>n</i>
Analytic			
Cluster 1	33.22	19.00	53
Cluster 2	59.29	22.09	47
Tone			
Cluster 1	15.74	14.95	53
Cluster 2	33.62	24.84	47
Clout			
Cluster 1	33.35	15.55	53
Cluster 2	68.25	15.93	47
Authentic			
Cluster 1	73.88	17.49	53
Cluster 2	38.52	21.35	47

Note. Listed means are the average of the standardized score for each summary variable.

Linear Discriminant Analysis. LDA was conducted on the four variables used to identify the clusters to determine if the clusters show separation in the data (Jombert et al., 2010). The LDA calculates components that are linear combinations of the four variables and exhibit the most separation in the data. Table 2 shows the results of the LDA, and Figure 2 shows the plot of the LDA for the components that explain at least 5% of the percentage of trace; with “trace” being the separation achieved by the discriminant function. Table 3 shows the

classification results of the LDA, demonstrating that 100% of cases were correctly classified. Figure 4 is a scatterplot of the cases assigned.

Table 2

LDA Cluster Means

Variable	<i>M</i>	<i>SD</i>	<i>n</i>
LD			
Cluster 1	-1.50	1.07	53
Cluster 2	1.69	0.91	47

Table 3

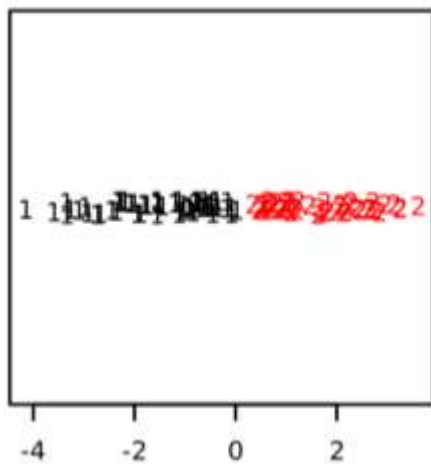
LDA Classification Results

		<i>Predicted Group Membership</i>			
		Summary Cluster Number	1	2	Total
Original	Count	1	47	0	47
		2	0	53	53
	%	1	100	0	100
		2	0	100	100
Cross Validated	Count	1	47	0	47
		2	0	53	53
	%	1	100	0	100
		2	0	100	100

Note. 100% of original grouped cases correctly classified. Cross validation was done only for cases in the analysis. During cross validation, each case as classified by the functions derived from all cases other than that case. 100% of cross-validated grouped cases were correctly classified.

Figure 4

LDA Plotted Along the x-Axis by Cluster



Every variable included in the LDA has a coefficient, which are the values used for the linear combinations to obtain the linear discriminant components. Correlations were also calculated to determine which variables had a large ($r = |.50|$) or moderate ($r = |.30|$) contribution to the LDA components (Cohen, 1988). For this LDA, *Analytic* ($r = .64$), *Clout* ($r = .88$), and *Authentic* ($r = -.80$) had a strong contribution and *Tone* ($r = .48$) had a moderate influence on the component. The variables that have large correlations for components, exhibiting a significant percentage of trace, contribute the most in separating the clusters. The standardized coefficients and the Pearson correlations can be seen in Table 4.

Table 4

Standardized Variable Coefficients and Pearson Correlations for Each Variable

Variable	Beta	<i>r</i>
Analytic	0.03	0.64
Tone	0.03	0.48
Clout	0.03	0.88
Authentic	-0.02	-0.80

Cluster 1 and Cluster 2 readily show good separation, and each variable demonstrates sufficient contribution to the formation of the clusters. The cluster formations align with the

theoretical notion of affective and instrumental offenders being distinguishable by linguistic style. Perpetrators in Cluster 1, what will now be called the *Affective Cluster*, appear to be more affectively driven. This cluster's linguistic markers show lower scores for Analytic, Clout, and Tone, and higher scores for Authentic. Cluster 2, or what will now be called the *Instrumental Cluster*, is identifiable by higher scores for Analytic, Clout, and Tone, with lower scores for Authentic. Figure 5 shows the score differences between the two clusters in profile plot with average summary variable scores displayed for each cluster. Figures 6 and 7 display box plots for the summary variable scores, with Figure 6 displaying Cluster 1 (Affective) and Figure 7 displaying Cluster 2 (Instrumental).

Figure 5

Profile Plot of Summary Variable Scores Grouped by Cluster

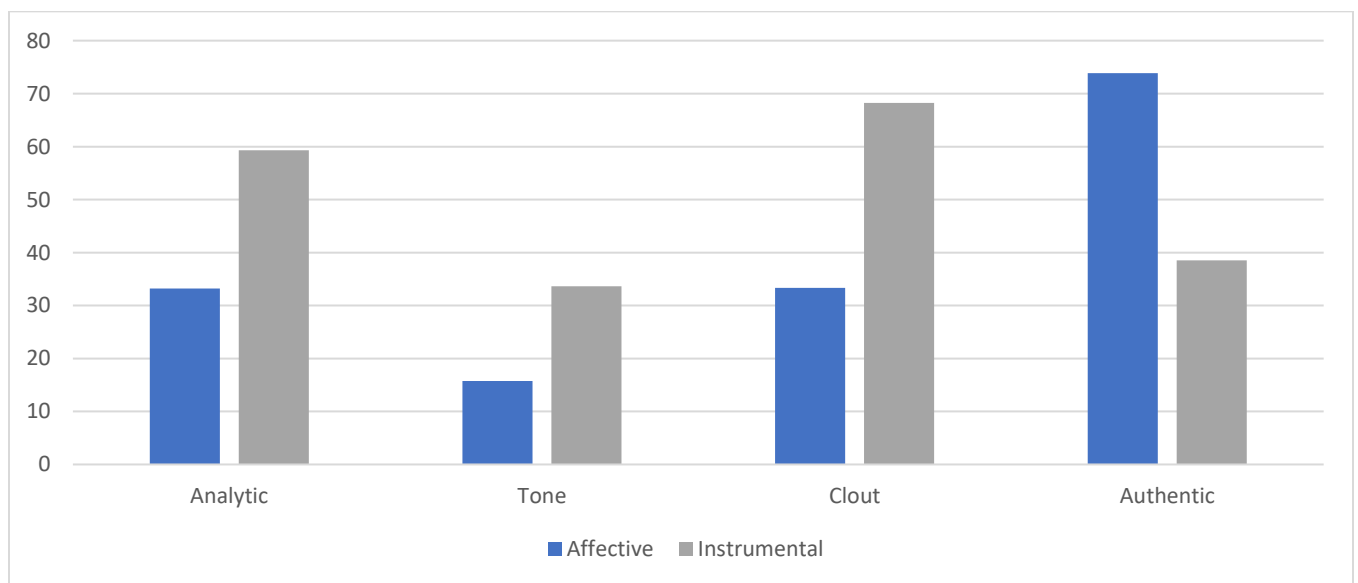


Figure 6

Box Plot of Summary Variable Scores for Cluster 1 (Affective)

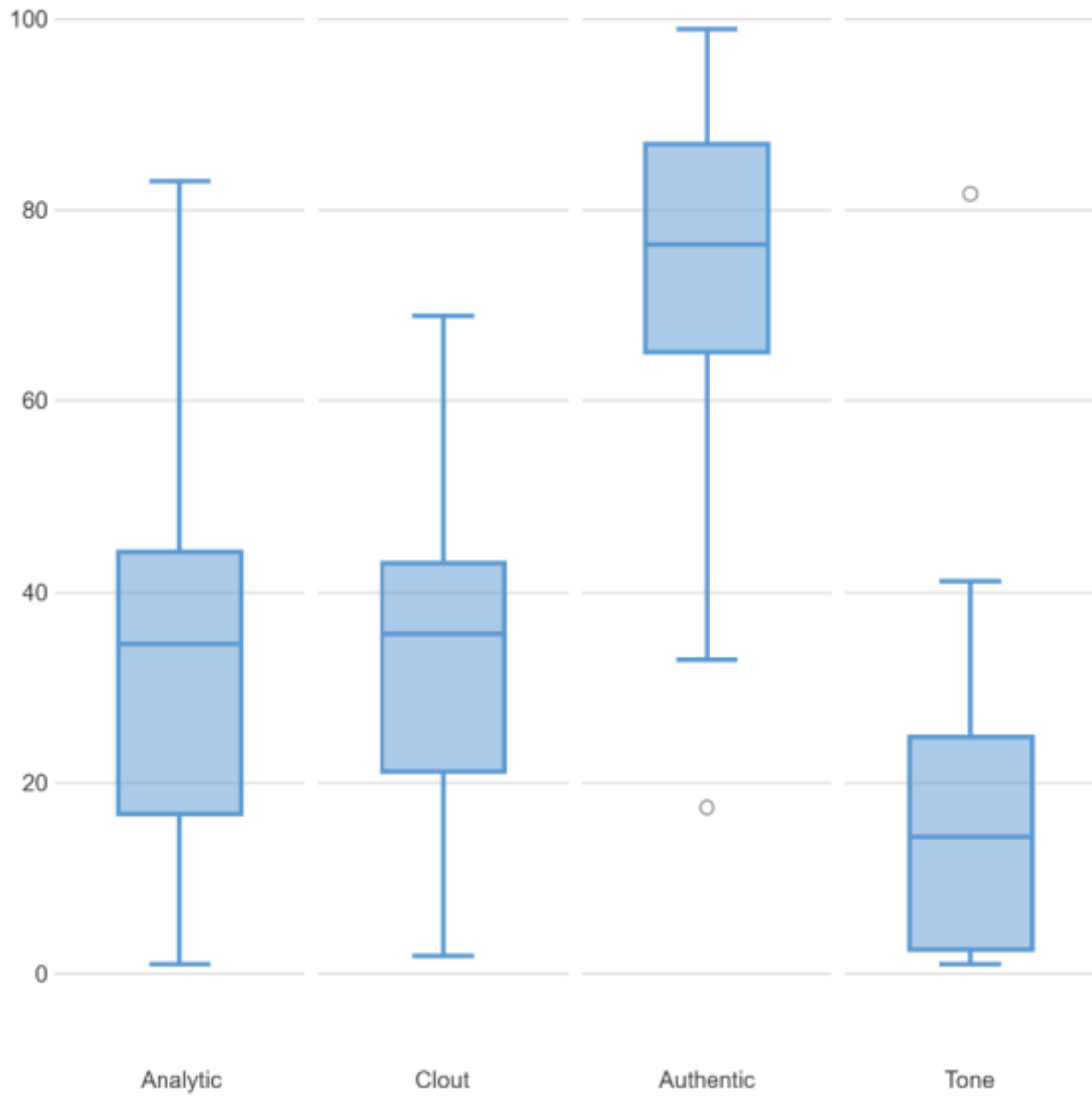
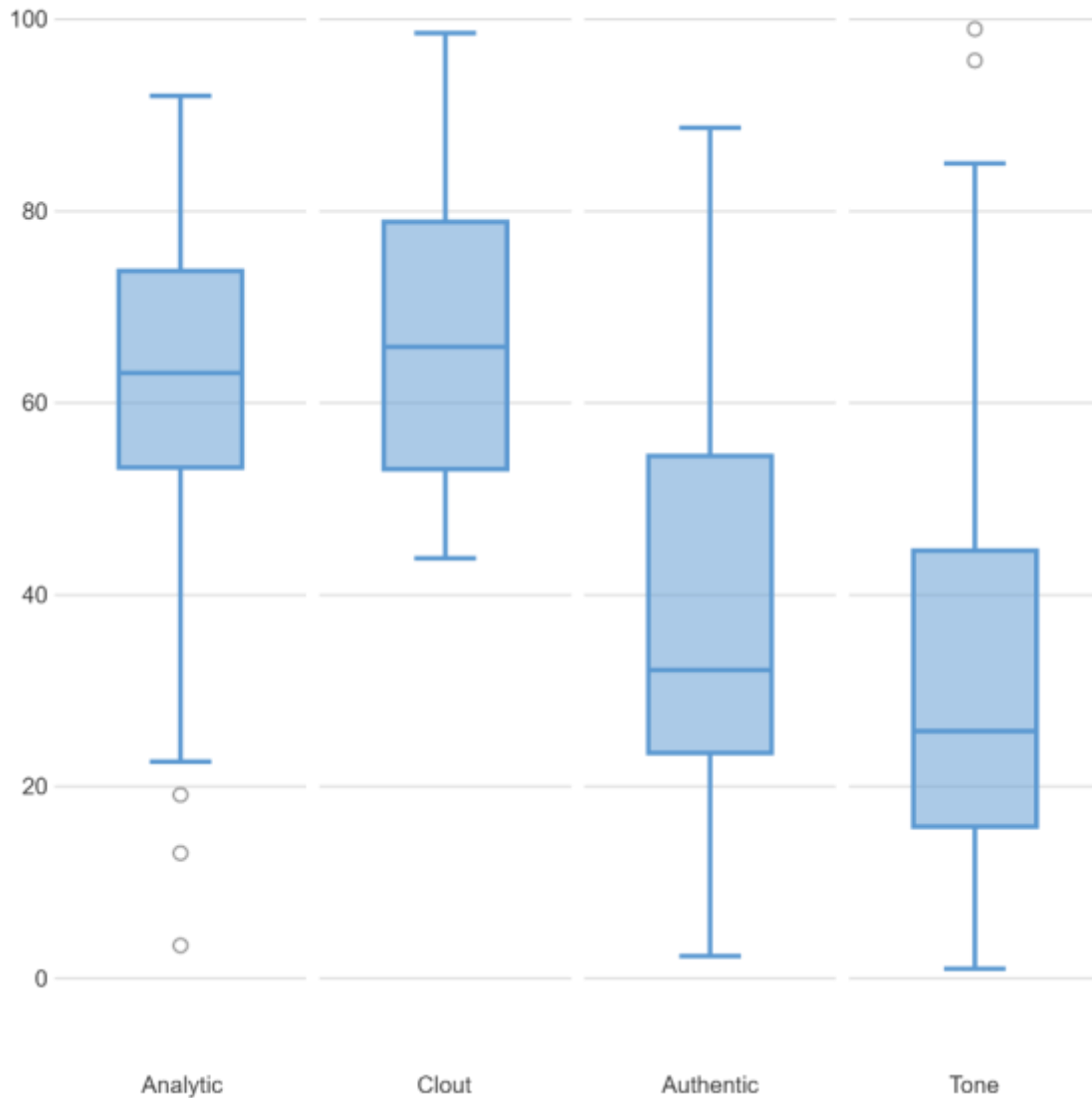


Figure 7

Box Plot of Summary Variable Scores for Cluster 2 (Instrumental)



Hypothesis 2: Cluster Membership Impact on Number of Victims Killed

The second hypothesis tested was that cluster membership based on summary language variables would have a statistically significant relationship with the number of victims killed. Prior to testing this hypothesis, the characteristics of the data were examined, and summary statistics can be found in Table 5. The histograms for number of victims killed for each cluster can be found in Figure 8 and Figure 9. Table 5 shows an observed skewness greater than two in

absolute value for the Instrumental Cluster, indicating asymmetry. Further, kurtosis is also greater than three for this cluster, suggesting a distribution markedly different from normal (Westfall & Henning, 2013).

Table 5

Summary Statistics Table for Victims Killed Separated by Cluster

Variable	<i>M</i>	<i>SD</i>	<i>n</i>	<i>SE_M</i>	Min	Max	Skewness	Kurtosis
Victims Killed								
Affective	4.55	4.56	53	0.63	0.00	17.00	1.09	0.03
Instrumental	12.38	15.81	47	2.31	0.00	77.00	2.08	4.87

Figure 8

Histogram of Number of Victims Killed Filtered by Affective Offenders

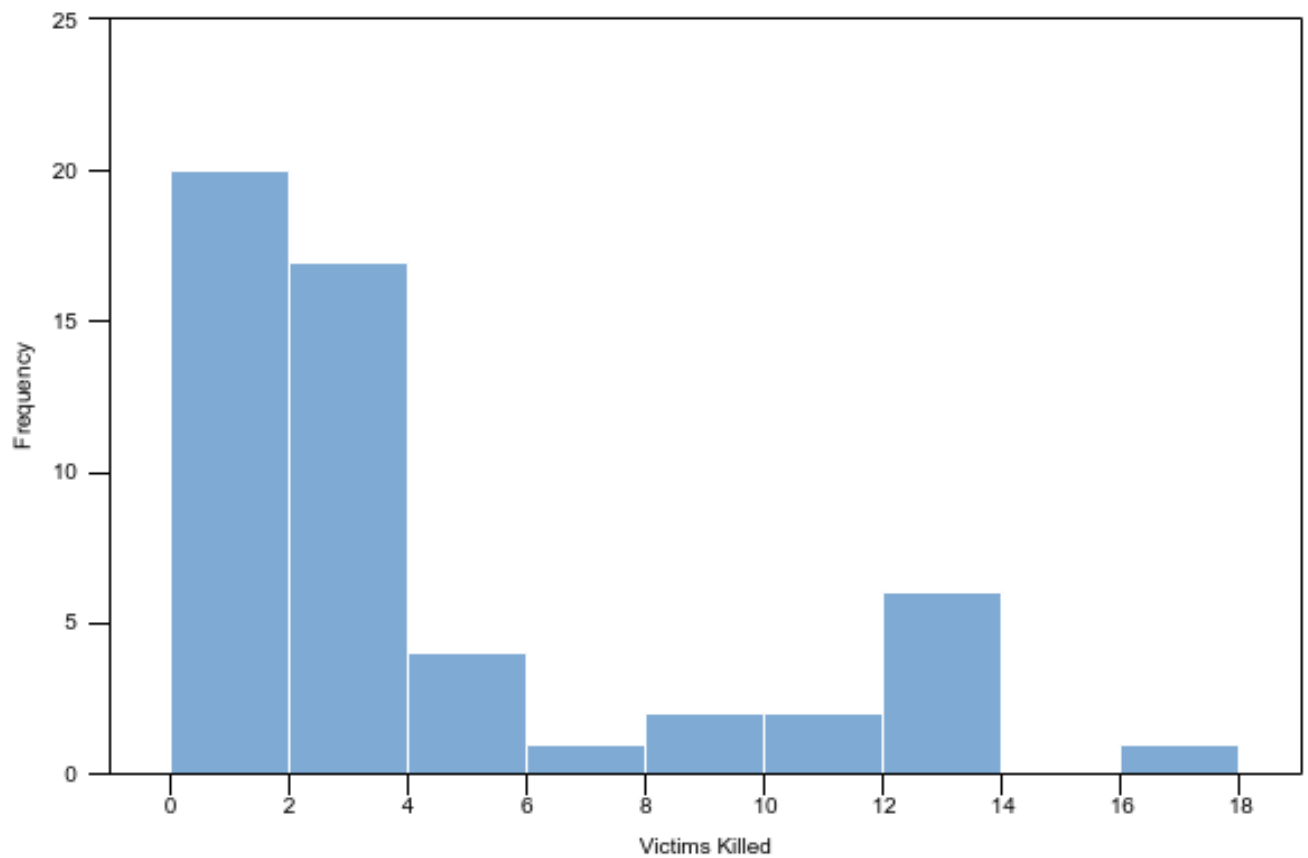
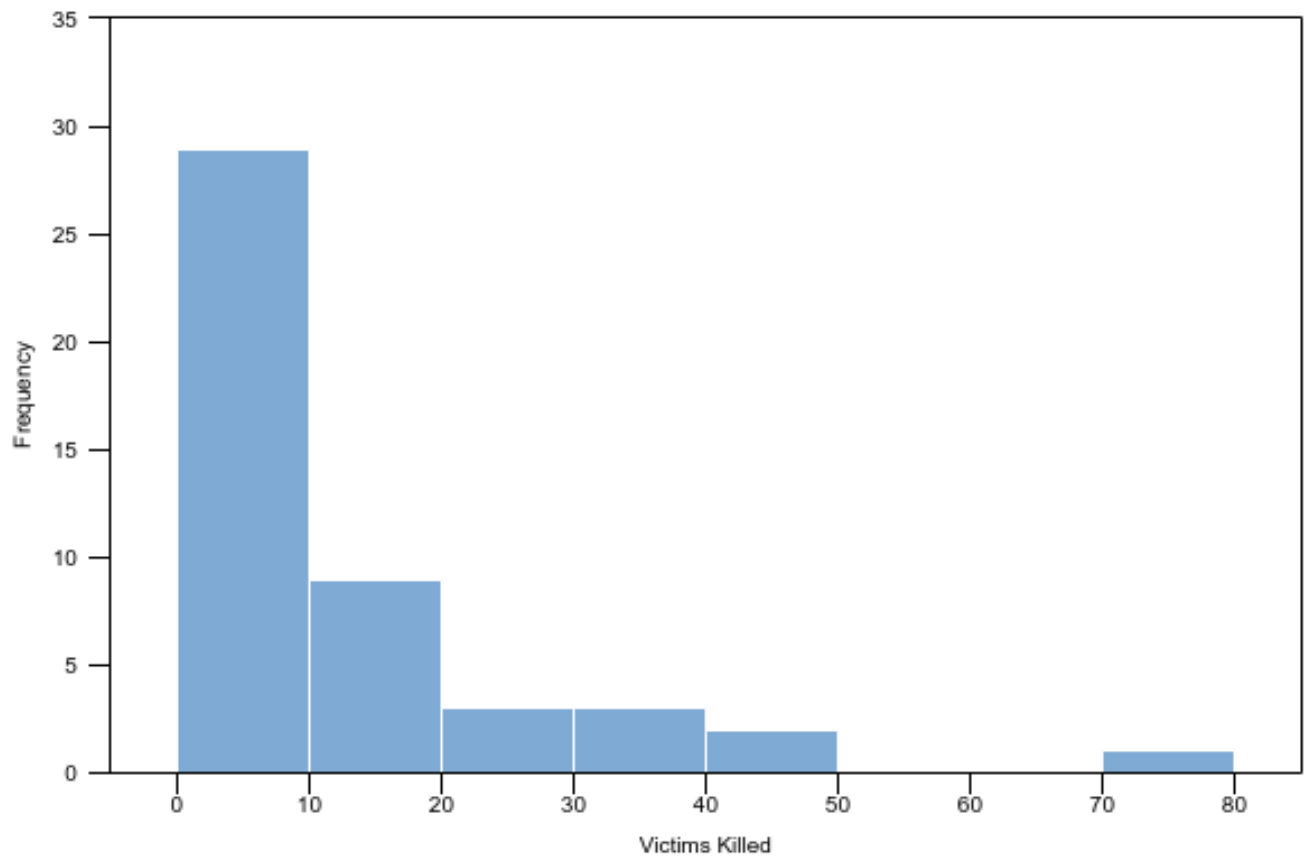


Figure 9

Histogram of Number of Victims Killed Filtered by Instrumental Offenders



The two-tailed Mann-Whitney U two-sample rank-sum test is an alternative to the independent samples t -test when assumptions for an independent samples t -test cannot be met (Conover & Iman, 1981). As the distributions between the two clusters were not normal but did share a similar shape, a two-tailed Mann-Whitney U two-sample rank-sum test was used to determine if the two clusters differed significantly in the number of victims killed. There were 53 observations in the Affective Cluster and 47 observations in the Instrumental Cluster.

The result of the two-tailed Mann-Whitney U test was significant based on an alpha value of .05, $U = 908$, $z = -2.35$, $p = .019$. The mean rank for the Affective Cluster was 44.13, and the mean rank for the Instrumental Cluster was 57.68. The results suggest that the distribution of the number of victims killed for the Affective Cluster was significantly different from the

Instrumental Cluster. Further, the median for the Affective Cluster ($Mdn = 3.00$) was significantly lower than the median for the Instrumental Cluster ($Mdn = 6.00$).

Hypotheses 3-5; Cluster Membership and Attack Behaviors:

Hypotheses three through five are that cluster membership will have a statistically significant relationship with whether an offender has a legitimate connection to the attack location (H_3), whether an attacker targeted specific individuals or types of people (H_4), and whether an attack ends with the perpetrator being killed (H_5). Fisher's exact test was used to test these three hypotheses.

For H_3 , the results of Fisher's exact test were significant based on an alpha value of .05, $p = .032$, suggesting that cluster membership and connection to the attack location are related to one another. This test was conducted with a 2 x 2 contingency table, and the odds ratio was calculated. $OR = 0.37$ [Inverted $OR (1/OR) = 2.7$], indicates that belonging to the Affective Cluster increases the odds that a perpetrator will be connected to the attack location by 270%, compared to those in the Instrumental Cluster.

For H_4 , the results of Fisher's exact test were not significant based on an alpha value of .05 ($OR = 2.09$, $p = .075$), suggesting that cluster membership and targeting could be independent of one another. This finding implies that the observed frequencies were not significantly different from the expected frequencies.

For H_5 , the results of Fisher's exact test were significant based on an alpha value of .05 ($p = .006$), suggesting that cluster membership and being killed at the conclusion of an attack are related to one another. $OR = 7.65$ indicates that the odds that a perpetrator will commit suicide or be apprehended rather than be killed by another person increases by 665% if the perpetrator is in the Affective Cluster. Table 6 presents the results of these tests, and Figure 10 displays the variable frequency as a percentage for the two clusters.

Table 5

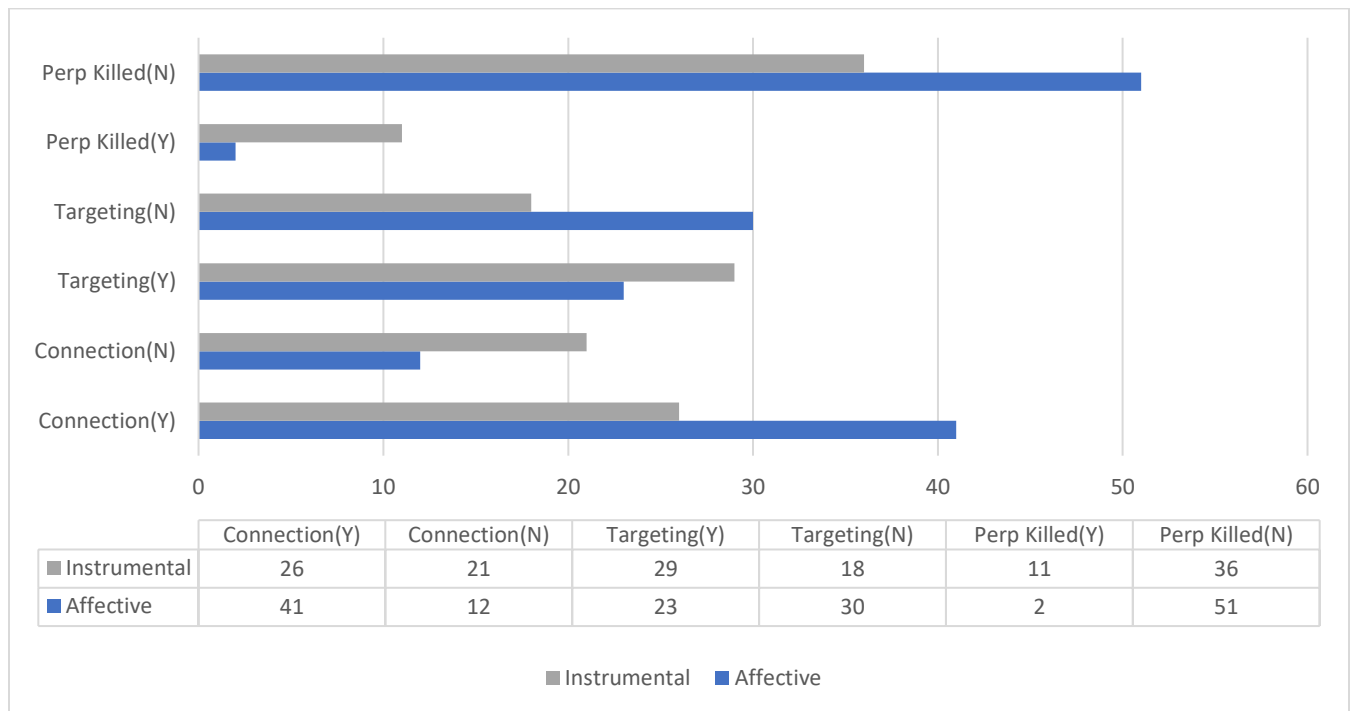
Fisher's Exact Tests of Variable Frequency Between Clusters for Connection, Targeting, and Perpetrator Killed

Variable	Cluster			<i>p</i>
	Affective	Instrumental	OR	
Connection				
No	12[17.49]	21[15.51]	0.37	.032
Yes	41[35.51]	26[31.49]		
Targeting				
No	30[25.44]	18[22.56]	2.09	.075
Yes	23[27.56]	29[24.44]		
Perpetrator Killed				
No	51[46.11]	36[40.89]	7.65	.006
Yes	2[6.89]	11[6.11]		

Note. Values formatted as Observed[Expected].

Figure 10

Variable Frequency Between Clusters for Connection, Targeting, and Perpetrator Killed



Note. (Y) = Yes; (N) = No

Hypothesis 6: Cluster Membership Modeling

The sixth hypothesis tested was that cluster membership could be modeled using binary logistic regression based on the number of victims killed, whether the attacker is personally connected to the attack location, whether the perpetrator engaged in victim targeting, and whether the attack ended with the perpetrator being killed. Even though binary logistic regression does not require the complete absence of multicollinearity, the presence of statistically significant relationships among the included variables was assessed before completing this analysis.

Two-tailed Mann-Whitney *U* tests were conducted to examine whether there were significant differences in the number of victims killed and whether the perpetrator had a legitimate connection to the attack location, and/or the presence of victim targeting, and/or whether the perpetrator was killed at the conclusion of the attack. None of these tests indicated significant differences in distribution based on an alpha value of .05. Table 7 presents the results of these tests.

Table 7

Two-Tailed Mann-Whitney U Tests for Number of Victims Killed

Variable	Mean Rank		<i>U</i>	<i>z</i>	<i>p</i>
	No	Yes			
Connection	53.88	48.84	1217.00	-0.82	.411
Targeting	48.00	52.81	1128.00	-0.83	.405
Perpetrator Killed	51.52	43.65	654.50	-0.92	.358

Fisher's exact tests were conducted to examine the relationship between connection to the attack location, targeting, and whether the perpetrator was killed. The results of Fisher's exact test examining the relationship between connection to the attack location and targeting were not significant based on an alpha value of .05 (OR = 1.03, *p* = 1.00), suggesting that connection to

the attack location and targeting could be independent of one another. This result implies that the observed frequencies and expected frequencies were not significantly different. Table 8 presents the results of this test.

Table 8

Fisher's Exact Test of Frequencies between Connection and Targeting

Targeting	Connection			<i>p</i>
	No	Yes	OR	
No	16[15.84]	32[32.16]	1.03	1.000
Yes	17[17.16]	35[34.84]		

Note. Values formatted as Observed[Expected].

The results of Fisher's exact test examining the relationship between connection to the attack location and whether the perpetrator was killed were significant based on an alpha value of .05 ($p = .027$, $OR = 0.26$, $1/OR = 3.85$), suggesting that connection to the attack location and the perpetrator being killed are related to one another. The test indicates that the odds of a perpetrator being killed are increased by 385% if there is no personal connection to the attack location. Table 9 presents the results of this test, and Figure 11 displays the frequencies as percentages for the four groups.

Table 9

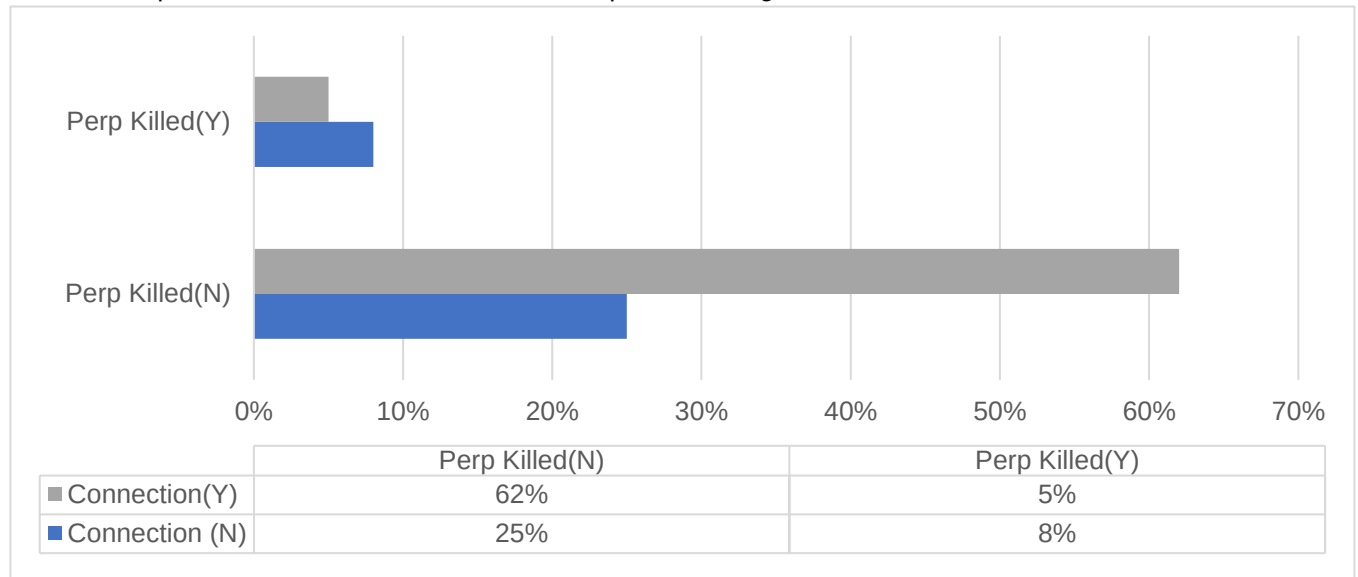
Fisher's Exact Test of Frequencies between Connection and Perpetrator being Killed

Perpetrator Killed	Connection			<i>p</i>
	No	Yes	OR	
No	25[28.71]	62[58.29]	0.26	.027
Yes	8[4.29]	5[8.71]		

Note. Values formatted as Observed[Expected].

Figure 11

Variable Frequencies Between Connection and Perpetrator being Killed



Note. (Y) = Yes; (N) = No

The results of Fisher's exact test examining the relationship between targeting and the perpetrator being killed were not significant based on an alpha value of .05 ($OR = 1.09$, $p = 1.00$), suggesting that the perpetrator's use of targeting and the perpetrator being killed at the scene of the attack could be independent of one another. This finding implies no significant difference between observed and expected frequencies. Table 10 presents the results of this test.

Table 10

Fisher's Exact Test of Frequencies between Targeting and the Perpetrator being Killed

Targeting	Perpetrator Killed		OR	<i>p</i>
	No	Yes		
No	42[41.76]	6[6.24]	1.09	1.000
Yes	45[45.24]	7[6.76]		

Note. Values formatted as Observed[Expected].

Following the completion of these tests, especially in consideration of the significant relationship between connection to the attack location and a perpetrator being killed, VIFs were calculated to detect the presence of problematic multicollinearity between the predictors. High VIFs indicate increased effects of multicollinearity in the model. VIFs greater than five are cause for concern, whereas VIFs of ten should be considered the maximum upper limit (Menard, 2009). All predictors in the regression model have VIFs less than two. Table 11 presents the VIF for each predictor in the model.

Table 11

Variance Inflation Factors for Perpetrator being Killed, Connection, Number of Victims Killed, and Targeting

Variable	VIF
Perpetrator Killed	1.24
Connected	1.10
No. of Victims Killed	1.16
Targeting	1.16

As there appeared to be no problematic multicollinearity, a binary logistic regression was completed. The resulting model was evaluated based on an alpha of .05. The overall model was significant ($\chi^2(4) = 30.58, p < .001$), suggesting that the perpetrator being killed, connection to the attack location, the number of victims killed, and the perpetrator's use of targeting had a collectively significant effect on the odds of observing each type of cluster membership. McFadden's R-squared was calculated to examine the model fit, where values greater than .2 are indicative of models with excellent fit (Louviere et al., 2000). The McFadden R-squared value calculated for this model was 0.22. In interpreting OR in this model, percentage of change to odds are calculated as $(OR - 1)100$. Additionally, the magnitude of all individual coefficients is interpreted as controlling for other predictors in the model.

The regression coefficient for the perpetrator being killed was significant ($B = 2.45, OR = 11.61, p = .010$), indicating that the odds of observing membership in the Instrumental Cluster increases by approximately 1,061% when a perpetrator is killed rather than commits suicide or is

apprehended. The regression coefficient for connection to the attack location was not significant ($B = -0.53$, $OR = 0.59$, $p = .315$), indicating that connection to the attack location did not significantly affect the odds of observing membership in the Instrumental Cluster or Affective Cluster. The regression coefficient for the number of victims killed was significant ($B = 0.12$, $OR = 1.12$, $p = .003$), indicating that the odds of membership in the Instrumental Cluster increases by approximately 12% for each victim killed. The regression coefficient for the perpetrator's use of targeting was significant ($B = 1.15$, $OR = 3.15$, $p = .023$), indicating that if a perpetrator targeted specific victims or types of victims, the odds of observing membership in the Instrumental Cluster increases by approximately 215%. Table 12 summarizes the results of the regression model.

Table 12

Logistic Regression Results with the Perpetrator being Killed, Connection, Number of Victims Killed, and Targeting Predicting Cluster Membership

Variable	B	SE	95% CI	χ^2	p	OR
(Intercept)	-1.44	0.64		5.15	.023	
Perpetrator Killed Yes	2.45	0.96	[1.78, 75.54]	6.58	.010	11.61
Connected Yes	-0.53	0.53	[0.21, 1.65]	1.01	.315	0.59
No. of Victims Killed	0.12	0.04	[1.04, 1.22]	8.79	.003	1.12
Targeting Yes	1.15	0.51	[1.17, 8.47]	5.15	.023	3.15

Note. $\chi^2(4) = 30.58$, $p < .001$, McFadden $R^2 = 0.22$.

The results of this model, though reasonable, are somewhat contrary to what the initial analyses suggested. Specifically, Fisher's exact test suggests that the perpetrator's use of targeting and cluster membership may be independent, but the binary logistic regression model shows that targeting has a significant impact on cluster membership. Further, Fisher's exact test also suggested that connection to the attack location and cluster membership are significantly related, but the connection variable was not found to contribute significantly to the model. This led the author to consider the potential presence of confounding variables and also to rethink the specificity and exclusivity of the variables in the dataset. Upon further thought, it was

determined that the perpetrator's use of targeting was poorly defined, as targeting was initially coded simply as *yes* or *no*, regardless of whether the perpetrator was targeting specific individuals or engaged in targeting based on demographic factors. Further, the specifier of *connection* suffered from similar problems, as it did not distinguish between whether the location that perpetrators were connected to was as intimate as their home or something less so, such as their place of employment or education.

It seems reasonable to consider that the perpetrator who attacks specific individuals differs from the perpetrator who only targets certain demographic groups and that there is also a difference between the perpetrator who kills at his home, potentially victimizing family members, and the perpetrator who chooses a location where a connection is still present but at a lesser degree. With these thoughts in mind, new variables were created to capture these different types of connection and targeting more specifically. New analyses were conducted, including the previously assessed variables of the number of victims killed and whether the perpetrator was killed, and the two new variables of whether the perpetrator engaged in demographic-based targeting and if an attacker's home was among the attack locations.

Mann-Whitney *U* tests were used to examine the differences between the number of victims killed, the presence of demographic-based targeting, and whether a perpetrator's home was among the attack locations; the number of victims killed and the perpetrator being killed was not performed a second time, as these variables were already found not to be significantly correlated.

The result of the two-tailed Mann-Whitney *U* test assessing the number of victims killed and a perpetrator's home being among the attack locations was significant based on an alpha value of .05 ($U = 616$, $z = -2.58$, $p = .010$). The mean rank for perpetrators whose home was not among attack locations was 46.21, and the mean rank for perpetrators whose home was among the attack locations was 63.36; suggesting that the distribution of the number of victims killed for perpetrators who did not include their home among their attack locations was significantly different from the distribution of the number of victims killed for perpetrators whose homes was an attack location. The median for the perpetrator's home not being among the attack locations

($Mdn = 3.00$) was significantly lower than the median for the perpetrator's home being among the attack locations ($Mdn = 6.00$).

The result of the two-tailed Mann-Whitney U test assessing the number of victims killed and whether the perpetrator engaged in demographic-based targeting was not significant based on an alpha value of .05 ($U = 1091$, $z = -1.10$, $p = .270$). The mean rank for the group not engaging demographic-based targeting was 47.32, and the mean rank for the group engaging demographic-based targeting was 53.68. These findings suggest that the distribution of the number of victims killed for those not engaged in demographic-based targeting ($Mdn = 3.50$) was not significantly different from the distribution of the number of victims killed for those who engaged in demographic-based targeting ($Mdn = 4.00$).

Fisher's exact test was used to look for significant relationships for the variables of the perpetrator being killed, engaging demographic-based targeting, and including one's home in the attack locations. The results of Fisher's exact test assessing the use of demographic-based targeting and a perpetrator's home being among the attack location were significant based on an alpha value of .05 ($p < .001$), suggesting that these two variables are related to one another. Table 13 presents the results of this test and shows that 50% of attackers did not engage in demographic-based targeting and their home was not among the attack locations, while the remaining 50% who did engage in demographic targeting were evenly split on whether their home was among the attack locations.

Table 13

Fisher's Exact Test of Frequencies between Demographic Targeting and the Perpetrator's Home being among the Attack Locations

Demographic Targeting	Perpetrator's Home		p
	No	Yes	
No	50[37.50]	0[12.50]	< .001
Yes	25[37.50]	25[12.50]	

Note. Values formatted as Observed[Expected].

The results of Fisher's exact test assessing the perpetrator being killed and the perpetrator's home being among the attack locations were not significant based on an alpha value of .05 (OR = 1.39, $p = .732$), suggesting that these variables could be independent of one another. This finding implies that the observed frequencies were not significantly different than the expected frequencies. Table 14 presents the results of this test.

Table 14

Fisher's Exact Test of Frequencies between the Perpetrator being Killed and the Perpetrator's Home being among the Attack Locations

Perpetrator Killed	Perpetrator's Home		OR	p
	No	Yes		
No	66[65.25]	21[21.75]	1.39	.732
Yes	9[9.75]	4[3.25]		

Note. Values formatted as Observed[Expected].

The results of Fisher's exact test assessing the perpetrator being killed and engaging in demographic-based targeting were not significant based on an alpha value of .05 (OR = 1.19, $p = 1.00$), suggesting that these variables could be independent of one another. Table 15 presents the results of this test.

Table 15

Fisher's Exact Test of Frequencies between the Perpetrator being Killed and Demographic Targeting

Demographic Targeting	Perpetrator Killed		OR	p
	No	Yes		
No	44[43.50]	6[6.50]	1.19	1.000
Yes	43[43.50]	7[6.50]		

Note. Values formatted as Observed[Expected].

As before, even though binary logistic regression does not require the complete absence of multicollinearity, these preliminary analyses suggest that there may be significant relationships among the contributing variables. To assess how problematic this may be, VIFs

were calculated. All predictors in the regression model have VIFs less than two. Table 16 presents the VIF for each predictor in the model.

Table 16

Variance Inflation Factors for Number of Victims Killed, Demographic Targeting, the Perpetrator's Home being among the Attack Locations, and the Perpetrator being Killed

Variable	VIF
No. of Victims Killed	1.15
Demographic Targeting	1.56
Perpetrator's Home	1.50
Perpetrator Killed	1.10

The binary logistic regression model, including the variables of the number of victims killed, the perpetrator's use of demographic-based targeting, the presence of the perpetrator's home among the attack locations, and whether or not the perpetrator was killed, was significant based on an alpha of .05 ($\chi^2(4) = 32.52, p < .001$), suggesting that the included variables had a collectively significant effect on the odds of observing each type of cluster membership. The McFadden R-squared value calculated for this model was 0.24, indicating a superior fit when compared to the originally examined model. As before, for the interpretation of the OR in this model, percentage of change to odds are calculated as $(OR - 1)100$ and the magnitude of all individual coefficients is interpreted as controlling for other predictors in the model.

The regression coefficient for the number of victims killed was significant ($B = 0.13$, $OR = 1.14, p < .001$), indicating that for each additional victim killed, the odds of observing membership in the Instrumental Cluster increases by approximately 14%. The regression coefficient for the use of demographic-based targeting was significant ($B = 1.57$, $OR = 4.83, p = .008$), indicating that when a perpetrator engaged in demographic-based targeting, the odds of observing membership in the Instrumental Cluster increases by approximately 383%. The regression coefficient for a perpetrator's home being among the attack locations was significant ($B = -1.62$, $OR = 0.20$ [Inverted $OR (1/.20 = 5)$], $p = .022$), indicating that if a perpetrator's home

is among the attack locations, the odds of observing membership in the Affective Cluster increases by approximately 400%. The regression coefficient for the perpetrator being killed was significant ($B = 2.65$, $OR = 14.11$, $p = .003$), indicating that if a perpetrator is killed, the odds of observing membership in the Instrumental Cluster increases by approximately 1,311%. Table 17 summarizes the results of the regression model.

Table 17

Logistic Regression Results with Number of Victims Killed, Demographic Targeting, the Perpetrator's Home being among the Attack Locations, and the Perpetrator being Killed Predicting Cluster Membership

Variable	<i>B</i>	<i>SE</i>	95% CI	χ^2	<i>p</i>	OR
(Intercept)	-1.72	0.47		13.64	< .001	
No. of Victims Killed	0.13	0.04	[1.06, 1.24]	11.03	< .001	1.14
Demographic Targeting Yes	1.57	0.59	[1.51, 15.38]	7.08	.008	4.83
Perpetrator's Home Yes	-1.62	0.70	[0.05, 0.79]	5.27	.022	0.20
Perpetrator Killed	2.65	0.90	[2.44, 81.65]	8.73	.003	14.11

Note. $\chi^2(4) = 32.52$, $p < .001$, McFadden $R^2 = 0.24$.

CHAPTER 6: DISCUSSION

Summary of Results

As described in the introduction to this paper, this project sought to explore the potential existence of an underlying, predictive psychological typology of individuals who commit mass murder. This project also aimed to investigate the utility of computer-mediated linguistic analysis to establish this typology and contribute to the field's understanding of the psychological characteristics of perpetrators of mass violence. The existing typologies initially presented to the reader can be classified into one of three categories: typologies that differentiate mass murderers from the general, non-offending population; typologies that distinguish mass murderers from other types of violent offenders; and typologies that seek to define different kinds of mass murderers. The analyses described in the preceding section add meaningfully to the last category, displaying substantial psychological differences between two types of mass murderers, with these differences being significantly related to specific attack features.

This present empirical study expands greatly on the purely theoretical typologies asserted historically by authors such as Dietz (1986), Holmes and Holmes (1994), and Kelleher (1997). Though others have previously attempted to use the above-mentioned theoretical models to inform empirical studies, as described in the literature review of this paper, those efforts created largely conflictual results, and little progress has been made in using these posited typologies as statistically significant predictors of certain attack behaviors. Generally speaking, contemporary literature only contains predictive models that differentiate attackers from the general population with some form of "likelihood for attack" metric, with no description of what kind of attack may occur. The analyses described in the above section represents substantial progress in the description of predictive factors that exist beyond simply whether a person is at risk for perpetrating or attempting to perpetrate an attack by adding critical threat assessment and risk

management information, such as possible location/victim types, potential lethality, and how an attack is more likely to end. Further, the preceding section also confirms the utility of computer-mediated linguistic analyses in typing perpetrators of mass violence and as something with encouraging potential for objective screening efforts after a pre-attack communication has been discovered and before an attack is underway.

The findings of this project reject what is presented as common wisdom and asserted by organizations such as the FBI's Behavioral Analysis Unit (2015); that all perpetrators of mass violence are best categorized as instrumental offenders. This study offers compelling evidence supporting the existence of two types of perpetrators of mass violence differentiated by objective, psychometric indications of affectivity. The current project also demonstrates a highly accurate classification method that categorizes perpetrators of mass violence as members of an Affective Cluster (1) or an Instrumental Cluster (2). Moving forward, these two types of perpetrators are referred to as the *Affective Type (1)* and the *Instrumental Type (2)*, with *Type* referring non-specifically to membership based on this model of categorization.

The Types revealed by this study showed clear separation based on LIWC2015 summary scores of Analytical Thinking (Analytic), Clout, Authentic, and Emotional Tone (Tone). As described earlier in this paper, a high score for Analytic reflects formal, logical, hierarchical thinking, with lower scores reflecting more informal, personal, present-focused, narrative thinking. High scores for Clout suggest the writer is communicating confident expertise, while lower numbers are associated with a writing style that is tentative, humble, and even anxious. Low scores for Authentic indicate a more guarded and/or distanced communication style, with high scores being generated by authors adopting a more honest and/or personal stance. Tone reflects emotional valence, with the higher numbers representing positivity, lower numbers

representing negativity, and numbers near 50 suggesting ambivalence or the absence of affective/emotive terms. In the analyses described in the prior chapter, Analytic, Clout, and Authentic strongly contribute to the cluster formation, while Tone exerts a moderate influence. The Instrumental Type of offender was found to achieve higher scores for Analytic, Clout, and Tone, with lower scores for Authentic, with the inverse being true of the Affective Type of offender.

Not only was Type clearly associated with psychometrically informative linguistic styles, Type was also found to have statistically significant associations with specific attack characteristics. Type was found to have a statistically significant relationship with the number of victims killed, whether the perpetrator has a connection to the attack location, and whether an attack is more likely to end with the perpetrator being killed by another person, as opposed to being taken into custody or committing suicide. The Affective Type was found to kill fewer people, be more likely to be connected to the attack location, and be less likely to be killed by another person at the conclusion of the attack. Though Type was not found to have a statistically significant relationship with whether victims were targeted or killed at random, this finding neared significance and may be worthy of additional investigation in the future ($p = 0.75$; Type I Error rate of 7.5% vs. 5%).

When binary regression modeling was used to examine the combined effect of attack characteristics on the prediction of cluster membership, the model was highly significant, with exceptional fit. Regression coefficients and odds ratios offered incredible findings. For instance, for each additional victim killed, the odds of the perpetrator being the Instrumental Type increases by approximately 14%. Additionally, if a perpetrator's home is among the attack locations, the odds of that perpetrator being of the Affective Type increases by 400%. Further,

when a perpetrator engages in demographics-based targeting, the odds of that perpetrator being of the Instrumental Type increases by more than 380%. Perhaps the most remarkable finding was that when an attack ends because the perpetrator was killed by another person, rather than committing suicide or being taken into custody, the odds of that perpetrator being of the Instrumental Type increases by more than 1,300%. Table 18 presents exemplars of each type, displaying summary language variable scores and observed attack behaviors for each exemplar.

Table 18

Type Exemplars

Name	Type	Analytic	Clout	Authentic	Tone	Victims Killed	Victims Injured	Connected to Attack Location	Outcome
Matthew Murray	Affective	18.81	46.47	89.35	1.00	4	5	Yes	Suicide
Anders Breivik	Instrumental	83.47	67.32	26.71	23.4	77	319	No	Apprehended

The Affective Type (1). Affective Type perpetrators kill fewer victims and are more likely to be personally connected to the attack location and to commit suicide at the conclusion of the attack. The linguistic markers of a Type 1 attacker include lower scores for the summary variables of Analytic, Clout, and Tone, and higher scores for Authentic. Qualitatively, the pre-attack communication of the Type 1 attacker is comparatively less formal, more personal, and hallmarked by present-tense, narrative thinking. This type of attacker’s communication is also relatively more tentative, humbler, and more likely to communicate anxiety, sadness, and hostility.

A Type 1 exemplar is Matthew Murray. In 2007, Murray fatally shot two people and wounded two others at a Christian missionary training center in Arvada, Colorado. Murray then traveled to a church in Colorado Springs, where he killed two more people and injured three

others before fatally shooting himself in the head. Prior to his commission of violence, Murray posted several communications online, some of which is presented in the following quote:

You christians brought this on yourselves. I'm coming for EVERYONE soon and I WILL be armed to the @#%\$ teeth and I WILL shoot to kill.God, I can't wait till I can kill you people. Feel no remorse, no sense of shame, I don't care if I live or die in the shoot-out. All I want to do is kill and injure as many of you ... as I can especially Christians who are to blame for most of the problems in the world. Well all you people out there can just kiss my (expletive removed) and die. From now on I don't give a @#%\$ about what all you (expletive removed) have to say, unless I respect you which is highly unlikely, but for those of you who do happen to know me and know that I respect you, may peace be with you and don't be in my line of fire, for the rest of you, you all better @#%\$ hide in your houses because I'm coming for EVERYONE soon, and I WILL be armed to the @#%\$ teeth, and I WILL shoot to kill and I WILL @#%\$ KILL EVERYTHING! No I am not crazy, crazy is just a word, to me it has no meaning, everyone is different, but most of you @#%\$ heads out there in society, going to your everyday @#%\$ jobs and doing your everyday routine (expletive removed) things, I say @#%\$ you and die, if you got a problem with my thoughts, come to me and I'll kill you, because... ..God (expletive removed), DEAD PEOPLE DON'T ARGUE! My belief is that if I say something, it goes. I am the law. If you don't like it, you die. If I don't like you or I don't like what you want me to do, then you die. If I do something incorrect, oh @#%\$ well, you die. Dead people can't do many things, like argue, whine, @#%\$, complain, name, rat out, criticize, or even @#%\$ talk. So that's the only way to solve arguments with all you (expletive removed) out there, I just kill. God I can't wait till I can kill you people, I'll just go to some downtown area in some big city and blow up and shoot everything I can. You break my back but you won't break me.....all is black but I still see...shut me down, knock me to the floor.....shoot me up, @#%\$ me like a whore....trapped under ice, comfortably cold, I've gone as low as you can go..... feel no remorse, no sorrow or shame....time's gonna wash away all pain I made a God out of blood not superiority I killed the king of deceit and now I sleep in anarchy.

Murray's posting shows a proliferative use of *I* and *you*, with a high level of emotionality apparent throughout. As mentioned in the literature review of this paper, greater use of first-person personal pronouns is associated with depression (Bernard et al., 2016; Himmelstein et al., 2018; Newell et al., 2018), suicidality (O'dea et al., 2017), physical and emotional pain (Tausczik & Pennebaker, 2010), general negative emotionality (Tackman et al., 2018), lower levels of social status and dominance (Arguello et al., 2006; Danescu-Niculescu-Mizil et al.,

2013; DeWall et al., 2011; Kacewicz et al., 2014; Pennebaker, 2011a), and as having an inverse relationship with measures of narcissism (Carey et al., 2015; Jones et al., 2017). Further, Murray states plainly, “I don’t care if I live or die...All I want to do is kill and injure as many of you as I can.” This statement reveals that, in accordance with the hypotheses present in this paper, as a Type 1, affectively motivated perpetrator, Murray viewed his act of violence as the end, not as a means to another end.

The Instrumental Type (2). Type 2 attackers can be characterized as the more instrumentally driven perpetrator. These perpetrators kill more victims and are less likely to be personally connected to the attack location. Type 2 perpetrators are also less likely to commit suicide at the conclusion of their attack, meaning that these attackers are likely to continue their violence until killed or apprehended. The linguistic markers of a Type 2 attacker include higher scores for the summary variables of Analytic, Clout, and Tone, with lower scores for Authentic. Given the theoretical and empirical implications of pronoun use, it would be reasonable to anticipate that these attackers would be less likely to use first-person singular pronouns (i.e., *I*), favoring first-person plural pronouns (i.e., *we*). Compared with the pre-attack communications of Type 1 perpetrators, the pre-attack communications of Type 2 perpetrators are comparatively more formal, logical, and reflective of hierarchical thinking. These perpetrators also appear to communicate from a perspective marked by apparent expertise and confidence. Further, the communications of these attackers are likely to exhibit a more guarded, distanced form of discourse while being more likely to be perceived as emotionally positive and/or upbeat.

A Type 2 exemplar is Anders Breivik. As mentioned previously in this paper, in 2011, Breivik killed 77 people and injured 319 others before being apprehended by authorities. In the

introduction to the web-published manifesto Breivik authored prior to his attack, Type 2 linguistic markers abound. Consider the following text:

The compendium/book presents advanced ideological, practical, tactical, organisational and rhetorical solutions and strategies for all patriotic-minded individuals/movements. The book will be of great interest to you whether you are a moderate or a more dedicated cultural conservative/nationalist.

Included are also demographical studies, historical statistics, forecasts and insights on various subjects related to the ongoing and future struggle of Europe. It covers most topics related to historical events and aspects of past and current Islamic Imperialism, which is now removed or falsified by our academia by instruction of Western Europe's cultural relativist elites (cultural relativism=cultural Marxism). It offers thorough analysis of Islam, which is unknown to a majority of Europeans. It documents how the political doctrines known as multiculturalism/cultural Marxism/cultural relativism was created and implemented. Multiculturalists/cultural Marxists usually operate under the disguise of humanism. A majority are anti-nationalists and want to deconstruct European identity, traditions, culture and even nation states.

As we all know, the root of Europe's problems is the lack of cultural self-confidence (nationalism). Most people are still terrified of nationalistic political doctrines thinking that if we ever embrace these principles again, new "Hitler's" will suddenly pop up and initiate global Armageddon... Needless to say; the growing numbers of nationalists in W. Europe are systematically being ridiculed, silenced and persecuted by the current cultural Marxist/multiculturalist political establishments. This has been a continuous ongoing process which started in 1945. This irrational fear of nationalistic doctrines is preventing us from stopping our own national/cultural suicide as the Islamic colonization is increasing annually. This book presents the only solutions to our current problems.

When contrasted with the writings of Murray, the differences are obvious. Despite the introduction to his manifesto being the most personal part of the entire document, Breivik's pronoun use is still markedly different, with him totaling avoiding the use of I. When Breivik uses pronouns, he primarily references a greater, forced affiliative group with *we*, *us*, and *our*, with a clearly delineated other/out-group. Further, throughout his introduction and the entirety of his manifesto, Breivik is highly dispassionate and seems to strive to convey the confident perspective of a scholarly expert.

Applications

Compared to other forms of violent offending, mass murder is a relatively rare occurrence; however, the impacts of this form of violence are wide-reaching and have a tremendous effect on the broader communal psyche. It is unknowable how much the relative rarity of this event is due to the prevention efforts enacted by threat assessment and risk management professionals; however, what is known is that these methods are imperfect and deserve improvement and enhancement through the work of social scientists. Although this study does not help determine whether an attack will occur, it represents a preliminary step for including linguistic analysis in the threat assessment and risk management process. If a pre-attack communication is discovered before a violent act is committed, the method outlined in this study could help determine what kind of attack is more likely to occur.

As discussed previously in this paper, most perpetrators display identifiable behaviors reflecting violent intent prior to the commission of these crimes (Fein et al., 1998; U.S. Secret Service & U.S. Department of Education, 2004). One of these behaviors is the creation of pre-attack communications. Assessors could use the typology presented in this study and the described methods of linguistic analysis to help identify what kind of threat is posed in a pre-attack communication and what prevention and mitigation efforts should be employed. Further, this analysis may aid responders in understanding if an attacker is likely to commit suicide or if the attack will continue until the perpetrator is killed or apprehended.

The primary benefit of the analysis presented in this research is that the methods employed could be modified into a process that is quantitative, automated, quickly replicable, and performable by laypersons. Consider a risk assessment team of a university that may be comprised of security officers, campus police, university administrators, and other university

staff. This team may be presented with what appears to be a pre-attack communication composed by a student. This team must now make decisions about the level of risk presented in the communication and what threat management strategies are most appropriate. This team could use a process informed by the current research to identify that the author of the pre-attack communication is best categorized as a Type 1 offender. This typing means that the author is likely to attack a location where a personal connection is present, such as the university itself, is less likely to kill a large number of victims and more likely to be in a highly emotional state and to commit suicide at the conclusion of the attack. Armed with this knowledge, the team could recommend target hardening of the university and employ an approach aimed at connecting the author of the suspected pre-attack communication with empathic and supportive mental health resources to reduce the overall level of risk and mitigate the presented threat. By acknowledging that Type 1 and Type 2 attackers essentially differ while being equipped with a method of differentiating these types reliably, threat management efforts can be better informed and more appropriately targeted.

Limitations

In accordance with Ethical Standard 9.01(b) of the American Psychological Association (2002), the author acknowledges without hesitation the limitations imposed by the lack of direct individual assessment. Further, despite completing exhaustive searches of all available data for the linguistic analysis of pre-attack communications, the relative rarity of this type of offense has caused this sample size to be less than ideal. Power analyses for chi-square tests were conducted in G*Power to determine a sufficient sample size using an α of .05, power of .80, and 1 df (Faul et al., 2013). For small, medium, and large anticipated effect sizes ($w = .1$, $w = .3$, and $w = .5$ respectively), for two categories for each assessed variable, the desired samples are 32, 88, and

785. For the analyses conducted with one variable with two categories and one variable with three categories, the desired sample sizes are 39, 108, and 964. The number of observations in this dataset is 100, which may have limited the measurement of moderate and small effect sizes and caused significance to have been underestimated. Additionally, unlike questionnaire item responses, words generally have low base rates and do not typically display normal distribution (Tausczik & Pennebaker, 2010). The underlying non-normality in the data led to the choice of using tests that are less sensitive to this data characteristic.

Unfortunately, there are some exclusions from this dataset because either no pre-attack communications exist, or they have not been made available to the public; such as in the case of Stephen Paddock, who in October of 2017 fatally shot 58 people and wounded 851 others at a concert in Las Vegas. Lastly, the relative recency of using computer-mediated linguistic analysis to make psychometric inferences and the novelty of using this method with this specific population limit the immediate practical application of the findings of this study. It is the author's intention for this research to be used as a starting point for further inquiry. It must also be noted that none of the findings of this research are intended to be predictive in terms of whether an attack will occur; rather, this analysis is aimed at determining what type of attack may occur based upon the examination of any discovered pre-attack communications.

Suggestions for Future Study

Due to the rarity of mass murder, the relatively small sample size limits the kind of analysis that can be performed and ability to fully observe all effects. As additional materials become available for analysis, the performed tests should be conducted again to see how additional observations impact the predictive quality of the typing system suggested in this paper. Further, researchers are encouraged to consider other attack behaviors not analyzed in the

current study that could inform the threat assessment and risk management process and other offender characteristics, such as age, that may contribute to cluster membership. Lastly, future research should be conducted to examine how the efficacy of specific prevention and mitigation efforts is influenced by the typing of a potential perpetrator.

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